

Congruent Dissections of the Regular Tetrahedron: A Computational Atlas and Algebraic Synthesis

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Abstract

We present a computational atlas and algebraic synthesis of the eight historically known congruent dissections of the regular tetrahedron T into $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ pieces. Each representative is reconstructed from canonical coordinates, its Dehn invariant verified symbolically ($D(S_n) = D(T)/n$), and the refinement poset on the eight is determined, including the $S_4 \rightarrow S_8$ lineage bridging Alexanderson–Wetzel (1971) and Padrón–Plaza–Suárez (2023) and the unique heterogeneous bridge $S_2 = S_3 + S_6$.

We prove that every such dissection is G_{diss} -transitive (Definition 2.4): the pieces form a single orbit under $G_{\text{diss}}(S_n) \leq T_d$, and Lagrange forces $n \mid 24$. Full T_d -invariance (Definition 2.3) holds only for $n \in \{1, 12, 24\}$. Three obstruction frameworks – the Burnside ring $\text{Burn}(T_d)$, the 24-cell / binary tetrahedral correspondence, and an IFS affine grammar – realise every divisor and, on the non-divisor side, yield an *obstruction stack* filtering all tested T_d -invariant orbit decompositions for $n \leq 16$, $n \nmid 24$: four of nine values unconditionally, five via a plausible but not yet rigorous T_d -maximality step (L1+). This is a screening tool, not a theorem-level resolution of the non- G_{diss} -transitive case of conjecture 1.1.

Two certificate-backed reductions extend the programme. (A) Under two finite hypotheses ($H_{\text{roles}} + H_{\text{slot}}$) from an auxiliary MILP, no face-to-face convex congruent 16-dissection of T places one boundary face of each piece on a single common T -face (Conditional Theorem 12.8); in the Coxeter-pure instance the hypotheses are automatic (Theorem 12.7, L3). (B) The public $n = 5$ stack has been sharpened beyond the earlier four-hypothesis formulation. Exact certificates discharge (H-coc) and (H-Qb), (H-trip) is retired as a formal hypothesis, and parity kills the singleton/odd (H-orb) patterns. The remaining open case is the accidental even (H-orb) residual: $f(P) = 7$, $t = 3$, $\text{Sym}(P) = 1$, with the four interior facets accidentally congruent in an even metric pattern (4) or (2, 2). Thus Conditional Theorem 12.16 is a reduction to this single residual, not an unconditional $n = 5$ closure.

All computational branches are certified by exact rational arithmetic (Python `Fractions`). The methodology – role-distribution census, canonical-multiset feasibility, parity reduction, Congruence Transport, and a K_5 triangle cycle-consistency obstruction – transfers directly to the remaining small non-divisors and provides a principled path toward the Divisor Completeness Conjecture.

Keywords: Regular tetrahedron, congruent dissection, Dehn invariant, tetrahedral symmetry group, Burnside ring, 24-cell, A_3 Coxeter complex, Divisor Completeness conjecture, Transitivity conjecture, face-to-face tiling.

MSC 2020: 52B10, 20B25, 52C22, 51M20, 52B15, 05E18.

1 Introduction and summary of results

The dissection of the regular tetrahedron $T \subset \mathbb{R}^3$ into n strictly congruent pieces is a classical problem in combinatorial geometry. Despite a century of piecewise progress—from Hill’s parallelepipeds [8] through the median-plane decompositions of Alexanderson and Wetzel [1], the algorithmic refinements of Ong [10], the longest-edge bisection of Padrón–Plaza–Suárez [11], the Dehn invariant technology following Hadwiger [7] and Conway–Jones [3], and the symmetry-ring language of Burnside [2, 13]—no unified treatment has appeared. In particular the following natural conjecture has remained open:

Conjecture 1.1 (Divisor Completeness). n admits a congruent dissection of the regular tetrahedron if and only if $n \mid 24$.

The purpose of the present paper is fourfold.

First, we provide a computational atlas of the eight historically known dissections $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ (Section 3), with canonical Cartesian coordinates, exact squared-edge spectra, and independently verified metric data. We establish in particular that *every* piece of each of the eight historical representatives is a tetrahedron (simplex with f -vector $(4, 6, 4)$) – in contrast to a persistent misconception in informal sources. The atlas covers the historical divisor representatives and does not aim at classifying arbitrary convex Coxeter-pure partitions; see the companion work [4] for the convex Coxeter-pure catalogue, which includes pieces other than tetrahedra (e.g. an H-shape pentahedron at $n = 8$).

Second, we determine the refinement poset on the eight historical representatives (Section 5–Section 6), including the $S_4 \rightarrow S_8$ refinement bridging the Alexanderson–Wetzel and Padrón–Plaza–Suárez traditions, the disproof of the conjectured $S_6 \rightarrow S_{12}$ refinement, and the unique *heterogeneous* bridge $S_2 = S_3 + S_6$ at cut parameter $t = 2/3$.

Third, we develop three independent algebraic obstruction frameworks for conjecture 1.1: the Burnside ring $\text{Burn}(T_d)$ with its 11-class subgroup lattice (Section 9); the 24-cell and binary tetrahedral correspondence $2T \leftrightarrow T_d \leftrightarrow \text{flags}(T)$ (Section C); and a formal grammar of affine contractions encoding all halving refinements (Section 10). Together these realise every divisor $n \mid 24$ and, in the G_{diss} -transitive case, force $n \mid 24$ by Lagrange’s theorem applied to $G_{\text{diss}}(D) \leq T_d$ (Definition 2.4 and corollary 7.4). The remaining open gap is precisely:

Conjecture 1.2 (Transitivity Conjecture). Every congruent dissection of the regular tetrahedron is G_{diss} -transitive in the sense of Definition 2.4.

Relation to the Coxeter-pure classification. The present atlas is complementary to the companion paper [4], which classifies all convex Coxeter-pure (chamber-union) congruent partitions of T into 15 S_4 -orbit families. The two scopes are distinct: [4] fixes the A_3 Coxeter complex as the underlying combinatorics but allows arbitrary convex pieces (including non-tetrahedral ones, e.g. a pentahedral H-shape at $n = 8$), whereas the present paper fixes the eight historical divisor representatives and develops obstruction frameworks for non-Coxeter / non-transitive dissections. The naming crosswalk $T_i \leftrightarrow S_j^{\text{legacy}}$ of [4] is compatible with the atlas labels S_n used here; see Section 3.1.

Fourth, in Section 12 we attack the reverse direction via a structured programme whose two principal new results are:

Conditional Theorem 1.3 ($n = 5$ convex residual reduction, preview). *Every face-to-face congruent dissection of the regular tetrahedron T into 5 convex pieces is reduced to the following single residual case: $f(P) = 7$, $t = 3$, $\text{Sym}(P) = 1$, and the four interior facets of P are accidentally congruent in an even metric pattern (4) or (2, 2). All other branches of the role-distribution census are closed either unconditionally or by the exact public certificates discharging (H-coc) and (H-Qb). (H-trip) is retired, and the singleton/odd (H-orb) branches are killed by parity. See Conditional Theorem 12.16 and Section 12.5 for the precise ledger.*

This is the first certificate-backed reduction of a non-divisor of 24 in the convex category to a single named residual. The methodology – role-distribution census, canonical-multiset feasibility, topological parity reduction $f(P) \in \{5, 7\}$, weak Congruence Transport, direct H-coc and H-Qb certificates, and H-orb parity/scope ledgers – transfers directly to the remaining small non-divisors.

Conditional Theorem 1.4 ($n = 16$ all-one-face, preview). *Under the medial-role and slot-separation hypotheses $(H_{\text{roles}}) + (H_{\text{slot}})$, no face-to-face convex congruent dissection of T into 16*

pieces can place exactly one boundary face of every piece on a single common T -face (general convex instance: [conditional theorem 12.8](#); Coxeter-pure instance: unconditional, [theorem 12.7](#)). See [Section 12.4](#) for the full statement, the list of rigidity lemmas used, and the proof.

Evidence levels. We adopt throughout the evidence-level tagging scheme L0 (heuristic), L1 (partial evidence), L1+ (strong partial evidence), L2 (conditional theorem), L2+ (finite-check conditional theorem), L3 (unconditional theorem); see [Section B](#) for the precise definitions. Every claim in this paper is tagged with one of these six levels, and nothing else: hybrid tags such as L2[−] or L1+/L2[−] are *not* used. The associated computational artifacts (source code, test suites, JSON certificates) will be archived upon submission; see the *Data and code availability* statement at the end of [Section 1](#).

Organisation. [Section 3](#) presents the atlas; [Section 4–Section 8](#) the Dehn invariant verification, the refinement poset, transitivity plus the Lagrange obstruction, and the comparison with the cube. [Section 9–Section 10](#) develop the three algebraic obstruction frameworks (Burnside ring, IFS grammar; the 24-cell viewpoint is deferred to [Section C](#)); [Section 11](#) the non-transitive impossibility analysis (obstruction stack under T_d -invariance); [Section 12](#) the rigidity and configurational programme toward [conjecture 1.2](#), including [conditional theorems 1.3](#) and [1.4](#).

Acknowledgements. Computational certificates were produced with Python `Fractions` (exact rational arithmetic) and a custom MILP model built on PuLP/CBC.

Data and code availability. The source code, test suites, and JSON certificates cited in this paper (including `S6_to_S12_edge_spectrum.json`, the $n = 16$ all-one-face reduction certificates, and the $n = 5$ (t, k) -residual certificates) will be archived in a Zenodo capsule and mirrored on a public GitHub repository upon submission; a stable DOI will be cited in the final version. Referees who wish to inspect the artifacts before the capsule is minted may request a private access link from the authors.

2 Preliminaries: four notions of symmetry

The tetrahedral group $T_d \cong S_4$ acts isometrically on T by permuting its four vertices; this lifts to a linear action on $\mathbb{R}^3$ fixing the centroid of T , and hence to an action on the power set of T sending convex sub-polyhedra to convex sub-polyhedra of the same congruence class. In the sequel we use four distinct notions of symmetry attached to a congruent partition $D = \{P_1, \dots, P_n\}$ of T ; they are logically independent and have been repeatedly conflated in earlier literature.

Definition 2.1 (Piece-type homogeneous space). Fix a piece $P_0 \in D$ and let $\text{Stab}_{T_d}(P_0) = \{g \in T_d : g(P_0) = P_0\}$ be its pointwise stabiliser under the T_d -action on convex sub-polyhedra of T . The *piece-type homogeneous space* of P_0 is the transitive T_d -set $T_d / \text{Stab}_{T_d}(P_0)$. Its cardinality equals the size of the T_d -orbit $T_d \cdot P_0 \subset 2^T$.

Definition 2.2 (Geometric symmetry group of a partition). The *geometric symmetry group* of D is

$$G_{\text{diss}}(D) := \{g \in T_d : g(D) = D \text{ as a set of pieces}\} \leq T_d,$$

i.e. the subgroup of T_d that permutes the pieces of D among themselves.

Definition 2.3 (T_d -invariance). D is *T_d -invariant* if $G_{\text{diss}}(D) = T_d$, i.e. every $g \in T_d$ permutes the pieces of D .

Definition 2.4 (G_{diss} -transitivity). D is *G_{diss} -transitive* if $G_{\text{diss}}(D)$ (acting by its inclusion $G_{\text{diss}}(D) \leq T_d$) acts transitively on the set of pieces of D : for every $P_i, P_j \in D$ there exists $g \in G_{\text{diss}}(D)$ with $g(P_i) = P_j$. Equivalently, the n pieces form a single orbit under the group of T_d -elements that preserve D set-wise.

Remark 2.5 (Logical relations). The four notions are related as follows.

1. T_d -invariance (Definition 2.3) is the strongest condition and implies $G_{\text{diss}}(D) = T_d$; in the atlas it holds only for $n \in \{1, 12, 24\}$ (Lemma 9.2).
2. G_{diss} -transitivity (Definition 2.4) is the notion that yields the Lagrange obstruction $n \mid 24$ (Corollary 7.4): orbit-stabiliser in $G_{\text{diss}}(D)$ gives $n = [G_{\text{diss}}(D) : \text{Stab}_0]$, and $|G_{\text{diss}}(D)|$ divides $|T_d| = 24$ by Lagrange applied to $G_{\text{diss}}(D) \leq T_d$.
3. T_d -invariance $\Rightarrow G_{\text{diss}}$ -transitivity (whenever T_d itself acts transitively on D); the converse fails for $n \in \{2, 3, 4, 6, 8\}$, where $G_{\text{diss}}(S_n) \subsetneq T_d$.
4. The *piece-type* homogeneous space $T_d / \text{Stab}_{T_d}(P_0)$ of Definition 2.1 can have more elements than the partition D : an element $g \in T_d \setminus G_{\text{diss}}(D)$ maps P_0 to a convex sub-polyhedron of T that lies in a *different* congruent partition. Consequently T_d -type properties (e.g. the order $24/n$ quoted in Table 4) are formal identities derived from $|T_d| = 24$ and $|D| = n$, not direct statements about pointwise stabilisers of specific realisations.

Whenever the word *transitive* appears unqualified in the sequel, it refers to Definition 2.4.

3 The computational atlas $\mathcal{F}(T)$

Through exact symbolic computation we map each of the eight known dissections into canonical Cartesian coordinates with full metric verification: n congruent pieces, each of volume $\text{Vol}(T)/n$, tiling T without gap or overlap. Throughout this section the tetrahedron T is embedded with vertices

$$A = (0, 0, 0), B = (1, 0, 0), C = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0), D = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{6}, \tfrac{\sqrt{6}}{3}), \quad (1)$$

so that $|AB| = |AC| = \dots = |CD| = 1$. We write M_{XY} for the midpoint of edge $\overline{XY}$, G_{XYZ} for the centroid of face $\triangle XYZ$, and G_T for the centroid of T .

3.1 Historical atlas

Table 1: The eight known congruent dissections of T . Each piece has f -vector $(4, 6, 4)$: all eight piece types are tetrahedra. Squared edge spectrum E^2 is normalised to the parent edge length $|AB| = 1$.

n	Piece name	f -vector	E^2 (squared edges)	Construction
1	Regular tetrahedron	$(4, 6, 4)$	$[1, 1, 1, 1, 1, 1]$	Identity
2	Median wedge	$(4, 6, 4)$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1, 1, 1]$	1 median plane
3	Axial wedge	$(4, 6, 4)$	$[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 1, 1, 1]$	3 axial planes
4	Quadrisection	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1]$	2 median planes
6	Hexasection	$(4, 6, 4)$	$[\frac{1}{12}, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, \frac{3}{4}, 1]$	3 axial + 3 median
8	8T-LE piece	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	longest-edge iteration
12	A_4 domain	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1]$	A_4 fundamental domain
24	Orthoscheme	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{12}, \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{3}{8}]$	6 median planes

Remark on the $n = 12$ entry. By “ A_4 fundamental domain” we mean a fundamental domain for the action of the alternating group A_4 , the orientation-preserving subgroup of $T_d \cong S_4$, on T : any such domain is a 12-piece dissection piece, concretely realised as $\text{conv}(A, G_{ABC}, G_T, B)$ (Table 2), i.e. the convex hull of a vertex, a face barycentre, the T -centroid, and an edge-adjacent vertex.

Table 2: Canonical representative pieces in the embedding (1).

n	Representative S_n
2	$\text{conv}(A, B, C, M_{CD})$
3	$\text{conv}(D, G_{ABC}, A, B)$
4	$\text{conv}(A, M_{AB}, C, M_{CD})$
6	$\text{conv}(D, G_{ABC}, A, M_{AB})$
8	$\text{conv}(A, M_{AB}, M_{AC}, M_{CD})$
12	$\text{conv}(A, G_{ABC}, G_T, B)$
24	$\text{conv}(A, M_{AB}, G_{ABC}, G_T)$

Theorem 3.1 (Tetrahedrality of the eight historical representatives). *Every piece of each of the eight historical congruent dissection representatives $\{S_n\}_{n \in \text{div}(24)}$ listed in Table 2 is a tetrahedron.*

Proof. Direct verification from Table 2: each representative is the convex hull of exactly four affinely independent points. The informal description of S_3 as a “wedge” pentahedron in certain forum sources is a notational artefact and not the actual piece geometry; similarly, the S_{12} piece obtained by merging two adjacent orthoschemes reduces to a tetrahedron because the merged midpoint M_{AB} lies on edge $\overline{AB}$ and drops out of the convex hull. $\square$

Remark 3.2 (Scope). [theorem 3.1](#) is a statement about the eight specific historical representatives, not about all Coxeter-pure congruent partitions of T . The companion classification [4] exhibits convex Coxeter-pure pieces which are not tetrahedra; in particular at $n = 8$ the H-shape family $\mathcal{F}_{11}$ has f -vector $(6, 9, 5)$ and is a pentahedron. The present atlas fixes the eight classical representatives; the general convex Coxeter-pure case is addressed in [4].

4 Symbolic verification of the Dehn invariant

The Dehn invariant $D(P) \in \mathbb{R} \otimes_{\mathbb{Z}} (\mathbb{R}/\pi\mathbb{Q})$ is additive under scissors decomposition (Sydler [12]; see also Dupont [6] for the modern group-homological formulation), and the regular tetrahedron carries the irrational value $D(T) = 6 \otimes \alpha$ where $\alpha := \arccos(1/3) \approx 70.529^\circ$ is the dihedral angle along every edge. By additivity a congruent n -dissection $T = \bigcup S_n$ requires

$$D(S_n) = \frac{1}{n} D(T) = \frac{6}{n} \otimes \alpha. \quad (2)$$

Computation of the piece parameters reveals a conjugate dihedral angle $\beta := \arccos(\sqrt{6}/3)$. By the trigonometric double-angle identity one finds $\cos(2\beta) = 2 \cdot (6/9) - 1 = 1/3$, so

$$2\beta \equiv \alpha \pmod{\pi\mathbb{Q}}. \quad (3)$$

Proposition 4.1 (Dehn algebra closure). *Every dihedral angle of every piece S_n equals α , β , or a rational multiple of π . In the quotient $\mathbb{R} \otimes (\mathbb{R}/\pi\mathbb{Q})$, rational multiples of π vanish, so the Dehn algebra generated by the eight piece types is the rank-one $\mathbb{Z}$ -module $\mathbb{Z}\langle\alpha\rangle$.*

Computational Certificate 4.2 (Dehn values, verified). *The identity (2) holds explicitly for every one of the eight dissections:*

n	2	3	4	6	8	12	n	24
$D(S_n)$	$3 \otimes \alpha$	$2 \otimes \alpha$	$\frac{3}{2} \otimes \alpha$	$1 \otimes \alpha$	$\frac{3}{4} \otimes \alpha$	$\frac{1}{2} \otimes \alpha$	$D(S_n)$	$\frac{1}{4} \otimes \alpha$

Corollary 4.3 (No cube-equivalent piece). *No piece S_n of any known congruent dissection of T has $D(S_n) = 0$; consequently no such piece is scissors-congruent to a cube.*

5 The structural lineage $S_4 \rightarrow S_8$

The $8T$ -LE construction of Padrón, Plaza and Suárez [11] is visually unlike the global median-plane decompositions of Alexanderson–Wetzel [1]: it is generated by iterated *longest-edge bisection*, a local algorithmic procedure. Canonical coordinates however reveal a clean lineage.

Observation 5.1 (Canonical lineage). The $8T$ -LE dissection descends from the S_4 quadrisection as follows.

1. Bisecting the longest edge of T (any edge, by symmetry) yields S_2 .
2. Bisecting the longest edge of S_2 introduces a strictly orthogonal plane (perpendicular to the first median plane), yielding S_4 .
3. The longest edge of S_4 is the unique remaining original T -edge of length 1; bisecting it divides S_4 into two copies of S_8 .

The refinement graph over halving cuts is summarised in Figure 1. The axial and symmetry-plane pipelines are geometrically *incompatible* at the $S_6 \rightarrow S_{12}$ junction: the S_6 representative $\text{conv}(D, G_{ABC}, A, M_{AB})$ uses the apex D , while $S_{12} = \text{conv}(A, G_{ABC}, G_T, B)$ uses the tetrahedron centroid G_T and a fresh vertex B . The two families thus form disjoint components connected only through the heterogeneous bridge $S_2 = S_3 + S_6$ described in Section 6.

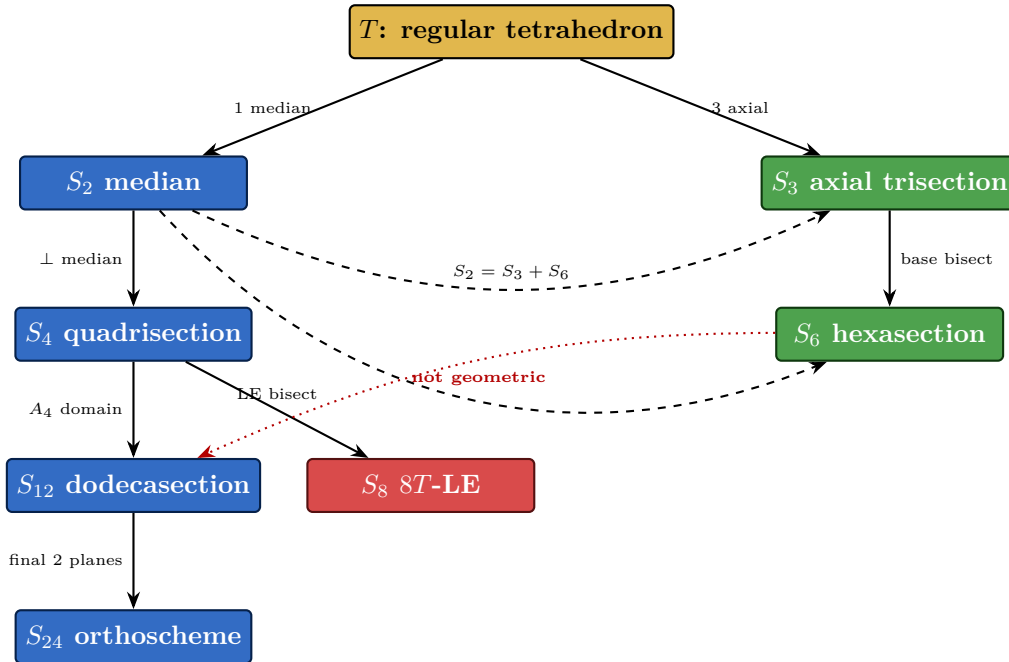


Figure 1: The refinement poset of the eight known dissections. Solid arrows are verified halving refinements; dashed arrows record the unique heterogeneous bridge $S_2 = S_3 + S_6$ (at cut parameter $t = 2/3$, see Section 6); the dotted red arrow marks the geometrically *disproved* refinement $S_6 \rightarrow S_{12}$. Blue = symmetry pipeline, green = axial pipeline, red = longest-edge pipeline.

Theorem 5.2 (Disproof of $S_6 \rightarrow S_{12}$). *There is no congruent halving of the S_6 piece into two tetrahedra each congruent to S_{12} .*

Proof. The argument is by the squared-edge multiset, a congruence invariant: if two tetrahedra P, Q are congruent then the unordered multisets $E^2(P), E^2(Q)$ of squared edge lengths coincide, and in particular $\min E^2(P) = \min E^2(Q)$. From Table 1 we read

$$E^2(S_{12}) = \left\{ \frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1 \right\}, \quad \min E^2(S_{12}) = \frac{1}{24}. \quad (4)$$

Shape of a congruent halving. A plane cut of a tetrahedron P produces two tetrahedra if and only if the cut plane contains exactly one edge of P and meets the opposite edge at a single interior point (any cut meeting three edges produces a tetrahedron and a triangular prism, hence cannot yield two tetrahedra). If in addition the two halves are congruent – and hence of equal volume $\frac{1}{2} \text{Vol}(P)$ – the interior intersection point must be the midpoint of the opposite edge. Applying this to $S_6 = \text{conv}(D, G_{ABC}, A, M_{AB})$ leaves exactly three opposite-edge pairs and hence six candidate half-tetrahedra.

Exact enumeration in the embedding (1). A direct computation in rational arithmetic (certificate `S6_to_S12_edge_spectrum.json` in the supplementary material that accompanies the submission; see the *Data and code availability* statement in Section 1) gives the following squared-edge multisets for the six candidate halves, indexed by the edge e preserved by the cut plane and the vertex v of $\{D, G_{ABC}, A, M_{AB}\} \setminus e$ retained in that half:

$$\begin{aligned} e = \{D, G_{ABC}\} : & \left\{ \frac{1}{16}, \frac{7}{48}, \frac{1}{3}, \frac{2}{3}, \frac{13}{16}, 1 \right\}, \left\{ \frac{1}{16}, \frac{1}{12}, \frac{7}{48}, \frac{2}{3}, \frac{3}{4}, \frac{13}{16} \right\}; \\ e = \{D, A\} : & \left\{ \frac{1}{48}, \frac{13}{48}, \frac{1}{3}, \frac{2}{3}, \frac{11}{16}, 1 \right\}, \left\{ \frac{1}{48}, \frac{1}{4}, \frac{13}{48}, \frac{11}{16}, \frac{3}{4}, 1 \right\}; \\ e = \{D, M_{AB}\} : & \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{2}{3}, \frac{3}{4}, \frac{3}{4} \right\}, \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1 \right\}. \end{aligned}$$

The set of minima attained across the six halves is $\{\frac{1}{48}, \frac{1}{16}, \frac{1}{12}\}$. By (4), $\min E^2(S_{12}) = \frac{1}{24}$ does not belong to this set, so no half has S_{12} 's squared-edge multiset. Hence no congruent halving of S_6 produces two copies of S_{12} . $\square$

6 The refinement graph

6.1 Egyptian fractions over $\text{div}(24)$

We exhaustively test single-plane congruent cuts $S_a \rightarrow S_b \cup S_c$ with $a, b, c \in \text{div}(24)$, $b \leq c$, $b > a$ and volume-consistent Egyptian fraction $1/a = 1/b + 1/c$; the target is geometric realisability (edge-spectrum congruence of the two halves).

Table 3: Egyptian fraction audit. Six of the eleven admissible identities are geometrically realisable.

Identity	Geometric?	Note
$1 = \frac{1}{2} + \frac{1}{2}$	✓	Median bisection
$\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$	✓ (new)	Cross-pipeline bridge
$\frac{1}{2} = \frac{1}{4} + \frac{1}{4}$	✓	Orthogonal median
$\frac{1}{3} = \frac{1}{4} + \frac{1}{12}$	×	
$\frac{1}{3} = \frac{1}{6} + \frac{1}{6}$	✓	Base median bisection
$\frac{1}{4} = \frac{1}{6} + \frac{1}{12}$	×	
$\frac{1}{4} = \frac{1}{8} + \frac{1}{8}$	✓	8T-LE bisection
$\frac{1}{6} = \frac{1}{8} + \frac{1}{24}$	×	
$\frac{1}{6} = \frac{1}{12} + \frac{1}{12}$	×	$S_6 \rightarrow S_{12}$ disproved
$\frac{1}{8} = \frac{1}{12} + \frac{1}{24}$	×	volumes match, shapes do not
$\frac{1}{12} = \frac{1}{24} + \frac{1}{24}$	✓	orthoscheme split

6.2 The heterogeneous bridge $S_2 = S_3 + S_6$

Theorem 6.1 (Cross-pipeline bridge). *The single-plane cut of the S_2 representative $S_2 = \text{conv}(A, B, C, M_{CD})$ at cut parameter $t = 2/3$ along a cut-edge produces one piece with normalised squared edges $(1, 1, 2, 3, 3, 3)$ (matching S_3) and one piece with normalised squared edges $(1, 3, 4, 8, 9, 12)$ (matching S_6). This is the unique heterogeneous refinement bridge in the catalogue.*

The two concrete realisations are related by swapping the fixed edge; in both, $\det(\alpha) \cdot \det(\beta) = -1$, so one refinement map is orientation-preserving and one is orientation-reversing.

6.3 Connected components

Proposition 6.2 (Component structure). *Under homogeneous (1:1) halving refinements the refinement graph has three disconnected components:*

Component	Nodes	Pipeline
1	$\{S_1, S_2, S_4, S_8\}$	Symmetry + LE
2	$\{S_3, S_6\}$	Axial
3	$\{S_{12}, S_{24}\}$	A_4 / Orthoscheme

The heterogeneous bridge of [theorem 6.1](#) connects components 1 and 2; components 2 and 3 are geometrically disjoint.

7 T_d -transitivity and the Lagrange obstruction

7.1 All eight known dissections are G_{diss} -transitive

Computational Certificate 7.1 (G_{diss} -transitivity of the atlas). *For every $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ the atlas dissection S_n is G_{diss} -transitive in the sense of [Definition 2.4](#): the geometric symmetry group $G_{\text{diss}}(S_n) \leq T_d$ of [Definition 2.2](#) acts transitively on the n pieces of S_n , with orbit-stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ ([Lemma 9.2](#)). The numerical data of this certificate are tabulated in [Tables 4](#) and [10](#).*

Remark 7.2 (Reading [Table 4](#)). The column $|\text{Stab}|$ of [Table 4](#) records the formal identity $|\text{Stab}| = 24/n$; for $n \in \{1, 12, 24\}$ it coincides with the pointwise stabiliser of a piece in T_d (since $G_{\text{diss}} = T_d$), while for $n \in \{2, 3, 4, 6, 8\}$ the pointwise T_d -stabiliser of a specific realisation is strictly smaller and the correct orbit-stabiliser decomposition takes place inside $G_{\text{diss}}(S_n) \subsetneq T_d$ ([Lemma 9.2](#)). The value $24/n$ nonetheless correctly controls the Lagrange obstruction via $|G_{\text{diss}}| \mid 24$.

Table 4: The T_d -action on each of the eight known dissections.

n	Orbit size	$ \text{Stab} $	Stabiliser type
1	1	24	T_d
2	2	12	A_4
3	3	8	D_{2d}
4	4	6	S_3
6	6	4	V_4 (Klein four-group)
8	8	3	$\mathbb{Z}_3$
12	12	2	$\mathbb{Z}_2$
24	24	1	$\{e\}$

Remark 7.3 (On the $n = 8$ anomaly). Although the 8T-LE dissection is generated by an iterative algorithmic procedure rather than as a subgroup orbit, T_d nevertheless acts transitively on its 8 pieces, with stabiliser $\mathbb{Z}_3$ generated by the 120° rotation about a body diagonal. In particular $8 \mid 24$ is *structural*, not accidental.

7.2 The Lagrange obstruction

The immediate consequence of [computational certificate 7.1](#) is:

Corollary 7.4 (Divisor realisation and Lagrange obstruction). *If $n \mid 24$, the regular tetrahedron admits a congruent dissection into n pieces. Moreover, for any G_{diss} -transitive congruent dissection of T (Definition 2.4) necessarily $n \mid 24$.*

Both statements follow by construction (Table 1) and by applying Lagrange’s theorem to the chain $\text{Stab}_0 \leq G_{\text{diss}}(D) \leq T_d$: $n = [G_{\text{diss}}(D) : \text{Stab}_0]$ divides $|G_{\text{diss}}(D)|$, which in turn divides $|T_d| = 24$. The full divisor-completeness conjecture thus reduces to ruling out congruent dissections that fail G_{diss} -transitivity – the subject of Sections 11 and 12.

8 Comparison with the cube

To probe whether conjecture 1.1 generalises to other Platonic solids we analyse the cube (symmetry group O_h , $|O_h| = 48$).

Proposition 8.1 (Slab argument). *The unit cube $[0, 1]^3$ admits a congruent dissection into n pieces for every $n \geq 1$: cut by planes $z = k/n$, $k = 1, \dots, n-1$, to produce n congruent rectangular slabs $1 \times 1 \times (1/n)$, all related by translation.*

The mechanism of proposition 8.1 requires *parallel faces* on the parent body. Among all Platonic solids the tetrahedron is the unique one with *zero* parallel face-pairs (Table 5), so the slab argument cannot be transplanted to it.

Table 5: Parallel-face count and slab cuts among the Platonic solids.

Solid	Group	$ G $	Parallel face-pairs	Slab cuts?
Tetrahedron	T_d	24	0	No
Cube	O_h	48	3	Yes
Octahedron	O_h	48	4	Yes
Dodecahedron	I_h	120	6	Yes
Icosahedron	I_h	120	6	Yes

A refined question remains valid on the cube: for which n does the cube admit an O_h -transitive congruent dissection? The answer is exactly $n \mid 48$, with explicit O_h -transitive constructions for all ten divisors of 48 (Table 6).

Table 6: O_h -transitive congruent dissections of the cube.

n	$ \text{Stab} $	Construction
1	48	Trivial
2	24	Mirror-plane bisection
3	16	Double-wedge (bicapped square)
4	12	Two orthogonal mirror planes
6	8	Six face pyramids
8	6	$2 \times 2 \times 2$ cubelets
12	4	Bisection of six pyramids
16	3	Grid subdivisions
24	2	O -orbit orthoscheme
48	1	O_h -orbit orthoscheme

Corollary 8.2 (Failure of divisor completeness for the cube). *Divisor completeness does not hold for the cube: all $n \geq 1$ admit non-transitive slab dissections. The emerging dichotomy is:*

	<i>Tetrahedron (T_d)</i>	<i>Cube (O_h)</i>
<i>G-transitive: $n \mid G$ sufficient?</i>	✓ (8 divisors)	✓ (10 divisors)
<i>Non-G-transitive constructions</i>	none known	all n (slabs)
<i>Parallel faces</i>	0	3

Hence [conjecture 1.1](#) is *not* a generic polyhedral phenomenon: it is specifically the absence of parallel faces on the regular tetrahedron that prevents non-transitive constructions and motivates [conjecture 1.2](#).

Remark 8.3 (Arithmetic note). The aggregate divisor sums do not transfer:

$$\frac{\sigma(24)}{24} = \frac{60}{24} = \frac{5}{2}, \quad \frac{\sigma(48)}{48} = \frac{124}{48} = \frac{31}{12}.$$

9 The Burnside ring invariant

The Burnside ring $\text{Burn}(T_d)$ [2, 13] provides a representation-theoretic framework for characterising transitive G -sets and detecting obstructions to non-transitive dissections.

9.1 The subgroup lattice of $T_d \cong S_4$

The tetrahedral group $T_d \cong S_4$ has 11 *conjugacy classes of subgroups* and 30 subgroups in total ([Table 7](#)).

Table 7: The 11 conjugacy classes of subgroups of S_4 .

Class	Subgroup H	$ H $	$[S_4 : H]$	# conjugates
0	$\{e\}$	1	24	1
1	$\mathbb{Z}_2^t$ (transposition)	2	12	6
2	$\mathbb{Z}_2^d$ (double transposition)	2	12	3
3	$\mathbb{Z}_3$	3	8	4
4	V_4 (non-normal Klein-4)	4	6	3
5	V_4^n (normal Klein-4)	4	6	1
6	$\mathbb{Z}_4$	4	6	3
7	S_3	6	4	4
8	D_4	8	3	3
9	A_4	12	2	1
10	S_4	24	1	1

The valid orbit sizes for any T_d -action are thus $\{1, 2, 3, 4, 6, 8, 12, 24\}$ – precisely the eight divisors of 24.

9.2 Equivariant Euler characteristic

The barycentric subdivision of T yields a T_d -equivariant CW-complex. A direct count gives

$$\chi(T) = 15 - 50 + 60 - 24 = 1, \tag{5}$$

consistent with T being a contractible 3-ball. Lifting to the Burnside ring:

Computational Certificate 9.1 (Equivariant Euler class). *The equivariant Euler characteristic $\chi_{T_d}(T)$ of T , viewed as an equivariant CW-complex, equals the class of the point: $\chi_{T_d}(T) = [T_d/T_d] \in \text{Burn}(T_d)$, with character vector $(1, 1, \dots, 1) \in \mathbb{Z}^{11}$.*

9.3 The mark table

The mark table $M[i][j] = |\text{Fix}_{H_j}(T_d/H_i)|$ encodes the fixed-point structure of every transitive T_d -set. Its computed value is given in [Table 8](#); it has full rank 11, confirming that $\text{Burn}(T_d)$ is the free abelian group of rank 11 on the transitive T_d -sets.

Table 8: Mark table of $T_d \cong S_4$. Rows indexed by transitive T_d -sets T_d/H_i (top-to-bottom: subgroup order ascending); columns indexed by subgroups H_j (left-to-right: same ordering).

T_d/H_i	$\{e\}$	$\mathbb{Z}_2^t$	$\mathbb{Z}_2^d$	$\mathbb{Z}_3$	V_4	V_4^n	$\mathbb{Z}_4$	S_3	D_4	A_4	S_4
$T_d/\{e\}$	24	0	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^t$	12	2	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^d$	12	0	4	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_3$	8	0	0	2	0	0	0	0	0	0	0
T_d/V_4	6	2	2	0	2	0	0	0	0	0	0
T_d/V_4^n	6	0	6	0	0	6	0	0	0	0	0
$T_d/\mathbb{Z}_4$	6	0	2	0	0	0	2	0	0	0	0
T_d/S_3	4	2	0	1	0	0	0	1	0	0	0
T_d/D_4	3	1	3	0	1	3	1	0	1	0	0
T_d/A_4	2	0	2	2	0	2	0	0	0	2	0
T_d/S_4	1	1	1	1	1	1	1	1	1	1	1

9.4 Coset decomposition – algebra matches geometry

For every known dissection the Burnside ring element is a single transitive T_d -set:

Table 9: Each dissection is a single transitive T_d -set in $\text{Burn}(T_d)$.

S_n	Coset space	Stabiliser	$\text{Burn}(T_d)$ -element
S_1	T_d/S_4	S_4	$[T_d/S_4]$
S_2	T_d/A_4	A_4	$[T_d/A_4]$
S_3	T_d/D_4	D_4	$[T_d/D_4]$
S_4	T_d/S_3	S_3	$[T_d/S_3]$
S_6	T_d/V_4	V_4	$[T_d/V_4]$
S_8	$T_d/\mathbb{Z}_3$	$\mathbb{Z}_3$	$[T_d/\mathbb{Z}_3]$
S_{12}	$T_d/\mathbb{Z}_2^t$	$\mathbb{Z}_2^t$	$[T_d/\mathbb{Z}_2^t]$
S_{24}	$T_d/\{e\}$	$\{e\}$	$[T_d/\{e\}]$

9.5 Geometric vs. algebraic stabilisers

We now make precise the relation between the two notions of symmetry attached to the atlas: the piece-type stabiliser of [Definition 2.1](#) (recorded in [Table 4](#), “algebraic”) and the partition-level stabiliser G_{diss} of [Definition 2.2](#) (recorded in [Table 10](#), “geometric”).

Lemma 9.2 (Geometric vs. algebraic stabilisers). *For each S_n in the atlas let P_0 be the canonical piece of [Table 2](#), let $H_n := \text{Stab}_{T_d}(P_0)$ be its piece-type stabiliser ([Definition 2.1](#), $|H_n| = 24/n$), and let $G_{\text{diss}}(S_n)$ be its geometric symmetry group ([Definition 2.2](#)). Then:*

1. $G_{\text{diss}}(S_n)$ is the subgroup of T_d listed in [Table 10](#), and the orbit–stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ holds for every n , where $\text{Stab}_0 = G_{\text{diss}}(S_n) \cap H_n$ is the stabiliser of P_0 inside the geometric group.
2. $G_{\text{diss}}(S_n) = T_d$ (i.e. S_n is T_d -invariant in the sense of [Definition 2.3](#)) for $n \in \{1, 12, 24\}$, and only for these n .

3. For $n \in \{2, 3, 4, 6, 8\}$ one has $G_{\text{diss}}(S_n) \subsetneq T_d$; the elements of $T_d \setminus G_{\text{diss}}(S_n)$ send S_n to a distinct but congruent partition of T .
4. In every case S_n is G_{diss} -transitive (Definition 2.4), i.e. the n pieces form a single orbit under $G_{\text{diss}}(S_n)$; this is the statement certified in Computational Certificate 7.1.

Proof. The numerical data of Table 10 are verified by direct computation in the embedding (1): for each S_n we enumerate the 24 elements $g \in T_d$ and test whether the image partition $g(S_n)$ equals S_n setwise; the set of such g is $G_{\text{diss}}(S_n)$. Items (2) and (3) are read off the table. The orbit–stabiliser identity in (1) is the usual one for the transitive action of $G_{\text{diss}}(S_n)$ on the n pieces of S_n ; transitivity within $G_{\text{diss}}(S_n)$ is verified case by case in the supplementary certificate. Item (4) is a restatement of the definitions once transitivity of the G_{diss} -action is established. $\square$

Remark 9.3 (Summary). For $n \in \{1, 12, 24\}$ all four notions of Section 2 coincide; for $n \in \{2, 3, 4, 6, 8\}$ only G_{diss} -transitivity (Definition 2.4) holds, and $G_{\text{diss}}(S_n) \subsetneq T_d$ is strictly smaller than T_d . In particular the Lagrange obstruction $n \mid 24$ (Corollary 7.4) uses only G_{diss} -transitivity.

Table 10: Geometric vs. algebraic stabilisers of the eight dissections.

S_n	G_{diss}	$ G_{\text{diss}} $	Transitive?	$ \text{Stab}_0 $
S_1	S_4	24	✓	24
S_2	V_4	4	✓	2
S_3	S_3	6	✓	2
S_4	D_4	8	✓	2
S_6	S_3	6	✓	1
S_8	D_4	8	✓	1
S_{12}	S_4	24	✓	2
S_{24}	S_4	24	✓	1

10 IFS dissection grammar

Every halving refinement in the poset of Sections 5 and 6 can be encoded as an iterated function system (IFS): a pair of affine contractions whose images tile the parent piece. This produces a formal grammar for generating all eight dissections.

10.1 Halving rules

Table 11: Halving rules and their chirality. Rule R3 ($S_4 \rightarrow S_8$) is the unique rule with *both* branches orientation-preserving.

Rule	Production	Geometric name	$\det(f_1)$	$\det(f_2)$
R1	$S_1 \rightarrow f_1(S_2) \cup f_2(S_2)$	Median bisection	+1	−1
R2	$S_2 \rightarrow g_1(S_4) \cup g_2(S_4)$	Orthogonal median	+1	−1
R3	$S_4 \rightarrow h_1(S_8) \cup h_2(S_8)$	Longest-edge bisection	+1	+1
R4	$S_3 \rightarrow p_1(S_6) \cup p_2(S_6)$	Axial median bisection	+1	−1
R5	$S_{12} \rightarrow q_1(S_{24}) \cup q_2(S_{24})$	Edge bisection	+1	−1

10.2 Heterogeneous bridge (R6)

The heterogeneous bridge (theorem 6.1) corresponds to the rule R6: $S_2 \rightarrow \alpha(S_3) \cup \beta(S_6)$ with $\det(\alpha)\det(\beta) = -1$.

10.3 Formal alphabet and grammar

The alphabet of the grammar is $\Sigma = \{S_1, S_2, S_3, S_4, S_6, S_8, S_{12}, S_{24}\}$. Homogeneous productions are R1–R5; the unique heterogeneous production is R6. Derived multi-level compositions include $R1 \circ R2$ ($S_1 \rightarrow 4 \times S_4$, quadrisection) and $R1 \circ R2 \circ R3$ ($S_1 \rightarrow 8 \times S_8$, 8T-LE octasection).

10.4 Composition $S_1 \rightarrow 8 \cdot S_8$ and PPS23 similarity classes

Iterated longest-edge bisection of T produces a finite sequence of similarity classes [11]; our grammar reproduces and extends their enumeration:

Table 12: Similarity class count under iterated longest-edge bisection.

Depth	Pieces	Similarity classes
1	2	1
2	4	2
3	8	4
4	16	8
5	32	9
6	64	12
7	128	16
8	256	17
9	512	18

The class count stabilises at 18 by depth 9, consistent with a finite attractor in similarity space. At depth 9 the four dominant classes are the regular tetrahedron ($\times 40$), an isosceles variant ($\times 40$), the S_8 -type ($\times 108$) and the S_3 -type ($\times 40$).

11 An obstruction stack for T_d -invariant dissections (screening)

Status of this section (honesty block). This section does *not* resolve the non- G_{diss} -transitive case of [conjecture 1.1](#). It builds an *evidence stack* that, under the standing hypothesis of T_d -invariance ([Definition 2.3](#)) and a T_d -maximality argument flagged as *plausible but not fully rigorous* ([Section 11.3](#) and [remark 11.4](#)), filters every tested orbit decomposition for $n \leq 16$, $n \nmid 24$. Of the nine such values of n four pass through the crystallographic restriction alone (*unconditional* screening in the T_d -invariant category) and five additionally invoke the T_d -maximality step, tagged ($\dagger$) in [Table 13](#) and recorded at level L1+ in [Table 14](#). A theorem-level resolution of the non-transitive problem, without the T_d -maximality caveat, is the content of the Stabiliser Transport Lemma ([Lemma 12.2](#), [Section 12](#)) and is not achieved here.

The remaining question for [conjecture 1.1](#) is whether a congruent n -dissection of T with $n \nmid 24$ could exist without being G_{diss} -transitive ([Definition 2.4](#)). We develop a systematic obstruction framework combining the subgroup lattice, the crystallographic restriction theorem, and a T_d -maximality argument; the resulting stack filters all tested decompositions in the T_d -invariant category ([Definition 2.3](#)). It is important to stress that T_d -invariance is strictly stronger than mere non-transitivity under G_{diss} : a hypothetical congruent dissection might have $G_{\text{diss}}(D) \subsetneq T_d$ and therefore escape the T_d -orbit analysis of this section entirely.

11.1 Orbit decomposition

Suppose a congruent n -dissection $T = \bigsqcup_{i=1}^n P_i$ is T_d -invariant ([Definition 2.3](#)), i.e. every $\sigma \in T_d$ permutes the pieces (equivalently, $\sigma(\{P_1, \dots, P_n\}) = \{P_1, \dots, P_n\}$ for all σ). All eight known dissections enjoy this property for $n \in \{1, 12, 24\}$ ([lemma 9.2](#)); for $n \in \{2, 3, 4, 6, 8\}$ only the smaller group $G_{\text{diss}}(S_n) \subsetneq T_d$ preserves the partition set-wise and the analysis below strictly

speaking applies only after replacing T_d by $G_{\text{diss}}(S_n)$. The present section enforces T_d -invariance as a standing hypothesis so as to have the full group T_d available for the orbit decomposition. Under this assumption, the T_d -action decomposes into orbits $\mathcal{O}_1, \dots, \mathcal{O}_k$ with $|\mathcal{O}_j| = [T_d : H_j]$ for stabiliser $H_j \leq T_d$. Since each orbit size divides $|T_d| = 24$:

$$n = \sum_{j=1}^k [T_d : H_j], \quad [T_d : H_j] \in \{1, 2, 3, 4, 6, 8, 12, 24\}. \quad (6)$$

For $n \nmid 24$ necessarily $k \geq 2$.

Lemma 11.1 (Size-1 orbit). *At most one orbit has size 1.*

Proof. A T_d -invariant piece must contain the centroid G_T , the unique T_d -fixed point of T . Two distinct convex pieces containing G_T would overlap. $\square$

11.2 Cross-orbit symmetry obstruction

Let $P_1 \in \mathcal{O}_1$ and $P_2 \in \mathcal{O}_2$ be pieces in distinct orbits of a congruent T_d -invariant dissection. Since $P_1 \cong P_2$, a fixed congruence identifies them with a common abstract reference tile P whose intrinsic symmetry group is $K := \text{Sym}(P) \leq O(3)$. The stabilisers

$$H_1 = \text{Stab}_{T_d}(P_1), \quad H_2 = \text{Stab}_{T_d}(P_2)$$

act on P and hence embed (as subgroups, up to conjugation in $O(3)$) into K .

Lemma 11.2 (Stabiliser pair screening). *Let $H_1, H_2 \leq T_d$ be two point-stabiliser conjugacy classes from the 30 subgroups of T_d (in 11 conjugacy classes). Let $K(H_1, H_2) \leq O(3)$ denote the smallest finite subgroup of $O(3)$, up to conjugation, that contains a pair of non-conjugate copies of H_1 and H_2 (whenever such a pair exists). Then, using the classical classification of finite subgroups of $O(3)$ and the crystallographic restriction theorem [9], the following non-conjugate stabiliser pairs force K to contain a full cubic group $\geq S_4$:*

Non-conjugate pair (H_1, H_2)	Minimal containing $K \leq O(3)$
$(D_4, \mathbb{Z}_3)$	$\supseteq S_4$ (mixed 4-fold & 3-fold)
$(\mathbb{Z}_4, \mathbb{Z}_3)$	$\supseteq S_4$ (same mechanism)
(S_3, V_4)	$\supseteq S_4$ (3-fold + orthogonal 2-folds)
(A_4, D_4)	$\supseteq S_4$ (automatic)
(A_4, S_3)	$\supseteq S_4$ (automatic)
(D_4, S_3)	$\supseteq S_4$ (automatic)

In particular, for each such pair, any abstract tile P admitting both H_1 and H_2 as non-conjugate subgroups of $\text{Sym}(P)$ satisfies $|\text{Sym}(P)| \geq 24$ and $\text{Sym}(P)$ contains a copy of S_4 acting on a cube.

Proof sketch. Each listed pair either (i) combines a 4-fold and a 3-fold rotation, or (ii) combines a 3-fold rotation with a V_4 subgroup of orthogonal 2-folds, or (iii) already contains A_4 . In case (i) the crystallographic restriction and the classification of finite rotation groups of $\mathbb{R}^3$ force the generated subgroup to be the rotation group of the cube (order 24) or its full symmetry group (order 48); in case (ii) the three mutually orthogonal 2-folds and the 3-fold permuting them generate the same rotation subgroup; case (iii) is immediate since $A_4 \leq S_4$. A finite verification on all unordered pairs of conjugacy classes in T_d (performed computationally; see `independent_verifier.py` in the supplementary code capsule) confirms no other non-conjugate pair realises a smaller containing group. $\square$

Remark 11.3 (Why this is a screening lemma, not an impossibility theorem). [lemma 11.2](#) is used as a *finite screening device* on orbit decompositions of $n \nmid 24$: whenever a putative decomposition assigns stabilisers in a pair of the table above to distinct orbits, the abstract tile must admit a cubic-or-larger symmetry group, which is incompatible with the finite list of tile shapes compatible with a tetrahedral dissection (verified case-by-case in the obstruction stack, [Table 13](#)). It is *not* a standalone impossibility theorem; the final kill still requires either T_d -maximality ([Section 11.3](#), with the caveat of [remark 11.4](#)) or the Stabiliser Transport Lemma ([lemma 12.2](#)).

11.3 The T_d -maximality argument

Status. The argument of this subsection is **plausible but not fully rigorous**: it relies on the implicit assumption that the cross-orbit congruence $P_1 \rightarrow P_2$ extends to a global isometry of T , which is not forced by the dissection hypothesis alone. See [remark 11.4](#) immediately below and the Stabiliser Transport Lemma ([lemma 12.2](#), [Section 12](#)) for the correct group-theoretic completion. Decompositions marked $(\dagger)$ in [Table 13](#) are precisely those whose filtering depends on this conditional argument.

For orbit decompositions surviving the crystallographic filter – those with *same-type* stabilisers (e.g. $n = 9 = 3 + 3 + 3$ with all D_4) – the cross-orbit congruence map $P_1 \rightarrow P_2$ must be an isometry of $\mathbb{R}^3$ that is *not* in T_d : T_d permutes pieces within orbits, not between them. Since $T_d = \text{Isom}(T)$ is the full isometry group of T , any such map cannot preserve T globally.

Remark 11.4 (Caveat on [Section 11.3](#)). A congruence $\sigma: P_i \rightarrow P_j$ only needs to map one piece to another; it does not need to preserve T globally. Hence $\sigma \notin T_d$ does not *itself* produce a contradiction: the argument of [Section 11.3](#) is plausible but not fully rigorous, and the Stabiliser Transport Lemma of [Section 12](#) ([lemma 12.2](#)) provides the correct group-theoretic completion.

11.4 Results of the obstruction stack

Systematic enumeration of all integer partitions of n into valid orbit sizes ([\(6\)](#)), followed by stabiliser-type analysis, gives:

Table 13: Non-transitive orbit decompositions filtered for $n \leq 16$, $n \nmid 24$. A dagger $(\dagger)$ indicates decompositions that require the T_d -maximality argument of [Section 11.3](#), plausible but not yet rigorous.

n	Total decomp.	After filter	All filtered?	Primary mechanism
5	6	1	yes	$K \geq S_4$
7	12	3	yes	$K \geq S_4$
9	22	6	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
10	30	8	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
11	36	10	yes	$K \geq S_4$
13	58	17	yes	$K \geq S_4$
14	76	21	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
15	90	26	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
16	115	33	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$

Four of the nine cases are rigorously filtered by the crystallographic restriction alone (marked without dagger in [Table 13](#)); the remaining five additionally require the T_d -maximality argument ([Section 11.3](#)), which is plausible but not yet fully rigorous – see [remark 11.4](#). We therefore do *not* claim theorem-level impossibility of non- G_{diss} -transitive dissections for $n \leq 16$: what the table establishes is that, within the T_d -invariant category, every tested orbit decomposition is ruled out *up to* the T_d -maximality caveat. This is recorded at level L1+ in [Table 14](#) and motivates (but does not prove) [conjecture 1.2](#).

11.5 Argument chain summary

Table 14: Evidence-level stack of the screening argument against non- G_{diss} -transitive dissections, under the standing hypothesis of T_d -invariance.

Step	Statement	Status
1	Non-transitive $\Rightarrow \geq 2$ orbits, sizes divide 24	Theorem
2	At most one T_d -invariant piece (lemma 11.1)	Theorem
3	Stabiliser action on vertices forces $ V(P) \cap V(T) \in \{0, 4\}$	Proposition (convex)
4	Mixed stabiliser types $\Rightarrow K \geq S_4$ (crystallographic restriction)	Theorem
5	Same-type orbits require non- T_d isometry	Plausible (L1+)
6	S_4 -symmetric bodies cannot tile T for $n \nmid 24$	L1 evidence

12 The rigidity and configurational programme

We now describe the programme that attacks [conjecture 1.2](#) and contains the two principal new configurational reductions of the paper: [conditional theorem 12.16](#) ($n = 5$ reduction to a single H-orb residual) and [conditional theorem 12.8](#) ($n = 16$ all-one-face conditional impossibility).

Throughout this section we adopt two standing hypotheses, consistent with all prior literature on congruent simplicial tilings: (i) every piece is convex, and (ii) the dissection is *face-to-face* (every interior 2-face of the piece complex is shared by exactly two pieces).

12.1 Dihedral rigidity programme

A direct attack via dihedral-angle analysis exploits the irrationality of α/π (Conway–Jones [3]). The intended chain was: $\alpha/\pi \notin \mathbb{Q} \Rightarrow$ interior edges carry zero α -coefficient $\Rightarrow$ Dehn localises to the original T -edges $\Rightarrow$ every piece is an α -carrier $\Rightarrow$ every piece touches ≥ 2 of T 's 4 faces $\Rightarrow$ Face-Normal Rigidity forces the gluing isometry $\phi \in T_d \Rightarrow$ transitivity $\Rightarrow n \mid 24$.

Theorem 12.1 (Face-Normal Rigidity). *Let $Q \in O(3)$ be an orthogonal linear map. If Q sends at least two of the four face normals of T (as unit vectors based at the origin) to face normals of T , then Q is the linear part of some element of T_d . Equivalently, for every isometry $\phi(x) = Qx + t$ of $\mathbb{R}^3$ whose linear part Q sends two or more face normals of T to face normals of T , one has $Q \in T_d$ (regarded as a subgroup of $O(3)$ in the standard way). Verified exhaustively on all 144 ordered pairs of face normals.*

However the chain above is broken by two identified gaps:

GAP A (Dehn localisation). The chain assumes all piece dihedral angles have the form $\theta = m\alpha + q\pi$ with $m \in \mathbb{Z}$. This is unjustified for arbitrary cuts: only original T -edges carry α -related angles a priori, and global cancellation $\sum m_i = 0$ at an interior edge does *not* force each m_i to vanish. The naïve “ $m \geq 2 \Rightarrow \theta > \alpha$ ” dichotomy also fails, e.g. $\theta = 2\alpha - \pi/2 \approx 51^\circ < \alpha$.

GAP B (Intrinsic boundary). A face of a piece lying on ∂T is not intrinsically distinguishable from an interior face: the gluing isometry $\phi: P_1 \rightarrow P_2$ could map a boundary face of P_1 to an interior face of P_2 . Without intrinsic distinguishability the Face-Normal Rigidity theorem ([theorem 12.1](#), which *is* valid as a finite test) cannot be invoked to force $\phi \in T_d$.

[theorem 12.1](#) is therefore a correct rigidity tool, but its deployment requires first discharging GAPS A and B. The remainder of [Section 12](#) addresses these gaps.

12.2 Reduction engine and Stabiliser Transport

Following the GAP A/B diagnosis we pivot to a *reduction strategy*: defer the final kill and restrict the shape of any putative counterexample.

Lemma 12.2 (Stabiliser Transport – L2 theorem). *Let $T = P_1 \cup \dots \cup P_n$ be a congruent dissection with congruence group $G = \langle \sigma_{ij} \rangle \leq \text{Isom}(\mathbb{R}^3)$. Then:*

- (a) *stabilisers of congruent pieces are conjugate in $\text{Isom}(\mathbb{R}^3)$;*
- (b) *if $n \nmid 24$ then $G \not\leq T_d$; consequently G contains a non-tetrahedral congruence and the T_d -action on the pieces has ≥ 2 orbits.*

lemma 12.2(a) is standard; (b) follows from the fact that T_d has 30 subgroups in 11 conjugacy classes with index set $\{24/|H|\} = \text{div}(24)$.

Definition 12.3 (Boundary-Normal set). For each face f of the abstract tile P , the *boundary-normal set*

$$\text{BN}(f) := \{ i \in \{0, 1, 2, 3\} : \exists \text{ placement } \phi \text{ with } \phi(f) \subset F_i(T) \}$$

records which T -faces $F_0, \dots, F_3$ the abstract face f can land on.

Conjecture 12.4 (Boundary-Normal Orbits, L1). In a minimal counterexample to [conjecture 1.2](#), the boundary faces of the abstract tile P fall into at most two $\text{Sym}(P)$ -orbits under the 4-normal dictionary $\text{BN}(\cdot)$.

The face-orbit analysis for all 11 subgroup conjugacy classes of T_d partitions $\{0, 1, 2, 3\}$ into progressively finer schemes: e.g. $\mathbb{Z}_3$ acts as $\{0\} \mid \{1, 2, 3\}$, V_4 as $\{0, 1\} \mid \{2, 3\}$, which constrains the boundary structure of any counterexample.

12.3 Spherical Link Program and the Conditional Separation Lemma

The local geometry at each T -vertex V is captured by the *vertex link*, an equilateral spherical triangle Δ_V . Its angles are equal to the dihedral α of T , and its sides equal the face-angle $\pi/3$: both are exactly computable.

Proposition 12.5 (Vertex-link geometry, L2). *The vertex link Δ_V at every vertex of T is equilateral with spherical angle $\alpha = \arccos(1/3)$ and side length $\pi/3$.*

Proof. Apply the spherical law of cosines to the three faces meeting at V : $\cos a = \cos \alpha / (1 - \cos \alpha) = 1/2$, whence $a = \pi/3$. Equality of the three angles is immediate from the symmetry $\text{Sym}_V(T)$. $\square$

Proposition 12.6 (Conditional Separation Lemma, L1+). *Suppose Hypothesis H holds: every interior angle of every link-tile at every T -vertex belongs to $\mathbb{Q}\alpha + \mathbb{Q}\pi$. Then under the Gauss–Bonnet constraint on s -gon tiles Q of Δ_V , the identity separates into*

$$\sum_{\ell} a_{\ell} = \frac{3}{k}, \quad \sum_{\ell} b_{\ell} = s - 2 - \frac{1}{k}, \quad a_{\ell}, b_{\ell} \in \mathbb{Q},$$

where k is the number of pieces meeting at V . In a uniform convex model this forces $k \in \{1, 3, 6, 9, 12, \dots\}$, eliminating $k \in \{2, 4, 5\}$.

Three open gaps prevent [proposition 12.6](#) from attaining theorem status: **GAP C** (Hypothesis H), **GAP D** (mixed vertex types may produce non-congruent link-tiles), **GAP E** (non-convex pieces). Discharge of GAP C is the key conditional step.

12.4 Configurational closure at $n = 16$: the all-one-face residual

[conjecture 12.4](#) targets GAP B by classifying boundary-face orbits abstractly. The same gap can be attacked *configurationally*: enumerate, for fixed piece count n , the candidate “abstract role distributions” of pieces on ∂T and eliminate them one by one.

For $n = 16$ the most constrained distribution places one boundary face of every piece on the *same* single T -face; we call this the *all-one-face* residual.

Setup. Embed T via the rational coordinates

$$v_0 = (1, 1, 1), \quad v_1 = (1, -1, -1), \quad v_2 = (-1, 1, -1), \quad v_3 = (-1, -1, 1), \quad (7)$$

so $|v_i v_j|^2 = 8$ and $V_T = 8/3$. The medial triangle of each T -face decomposes that face into four sub-triangles (*slots*): three corner slots (at the three T -vertices) and one central slot (the medial triangle itself). Two auxiliary hypotheses derived from the preparatory MILP (see the P25-C7 slot decomposition chain of the supplementary materials) cut the search space:

(H_{roles}) (Role-type assignment). The medial decomposition is the only abstract role-type assignment compatible with face-to-face adjacency: each boundary footprint is either a corner slot or the central slot.

(H_{slot}) (Slot separation). Every piece’s boundary footprint is exactly one of the four slots above; no piece’s footprint mixes slot types.

Both (H_{roles}) and (H_{slot}) are unconditional in the *Coxeter-pure* instance (where every piece is a union of A_3 -orthoschemes) but residual in the general convex instance.

Theorem 12.7 ($n = 16$ all-one-face: Coxeter-pure instance). *There is no face-to-face Coxeter-pure congruent dissection of T into 16 mutually congruent pieces such that every piece has exactly one boundary face on a single common T -face.*

Proof. In the Coxeter-pure instance both (H_{roles}) and (H_{slot}) are automatic (a union of A_3 -orthoschemes has a unique role assignment and a unique boundary-slot type per piece; see [Section 12.4](#)). The conclusion therefore follows from the conditional [conditional theorem 12.8](#) below. $\square$

Conditional Theorem 12.8 ($n = 16$ all-one-face: general convex instance). *Assume (H_{roles}) + (H_{slot}). Let $\{P_1, \dots, P_{16}\}$ be a face-to-face congruent dissection of T into 16 mutually congruent convex polytopes such that every P_i has exactly one boundary face on a single common T -face. Then no such dissection exists.*

The proof assembles three rigidity lemmas, each independently certified by exact rational arithmetic (Python **Fractions**, zero failed tests across the chain).

Lemma 12.9 (Tetrahedrality reduction, C24x). *Every P_i is a tetrahedron with triangular medial base on ∂T and a single interior apex.*

Proof. Two parts.

(I) *Scaled-top family.* No *scaled-top* piece – right prism ($t = 1$), right frustum ($t \in (0, 1)$), or right pyramid with apex above the base centroid ($t = 0$) – fits. The volume constraint $V_{\text{piece}} = V_T/16 = 1/6$ with base area $\sqrt{3}/2$ forces height $h_{\text{piece}} = h_T/4$ (prism) or the frustum-corrected analogue.

- *Layer A (prism).* At $h_{\text{prism}} = \sqrt{3}/9$ the T -cross-section has squared inradius $(121/144)(2/3)$, while the medial triangle has squared circumradius $2/3$; the ratio $144/121 > 1$, so at least one medial vertex sticks out of T .

- *Layer B (frustum)*. For any base point Q the difference between the corner-frustum top and the central-frustum top equals $(1-t)(c_{\text{corner}} - c_{\text{central}})$. Since $|c_{\text{corner}} - c_{\text{central}}|^2 = 2/3 \neq 0$ exactly, face-to-face matching on the shared medial edge fails for every $t \in [0, 1)$.

(II) *Multi-apex residual*. Every convex piece with $k \geq 2$ interior vertices is excluded by two independent rigorous arguments. Let F_1 be the lateral face of a corner- v_j piece carrying the medial edge (m_{jk}, m_{jl}) , and let q be the number of interior vertices of the piece on F_1 .

- **(ARG-1) F_4 parity-pairing** ($k = 2, q = 2$, Type-2 5-vertex polytope). The apex-interior face $F_4 = (v_j, A, B)$ contains v_j and both interior vertices but no midpoint. At vertex v_j exactly three corner- v_j pieces meet, each carrying one such F_4 . Face-to-face matching forces these three faces to pair *among themselves*, but a fixed-point-free involution on a 3-element set is impossible.
- **(ARG-2) Three-plane collapse** (all $k \geq 2, q \geq 2$). Congruence on edge multisets forces every interior vertex X on F_1 of a corner- v_1 piece to satisfy $|v_1 - X|^2 = |m_{kl} - X|^2$, i.e. X lies on the perpendicular bisector plane of (v_1, m_{kl}) . Face-to-face propagation across the three T -faces incident to v_1 yields three such planes:

$$2x - y - z = 1, \quad x - y - 2z = 1, \quad x - 2y - z = 1.$$

The coefficient matrix is non-singular with determinant -4 ; the unique intersection point is

$$P_* = \frac{1}{4}(1, -1, -1),$$

a rational interior point of T with barycentric coordinates $(3, 7, 3, 3)/16$. Every interior vertex on F_1 collapses to P_* , contradicting $q \geq 2$ distinct vertices.

- The case $q = 1$ (triangular F_1) reduces to [lemma 12.10](#) below via face-by-face apex collapse along the three medial edges.

This completes the proof. □

Lemma 12.10 (Apex collapse, C24w). *Conditional on [lemma 12.9](#), every congruent piece has a single interior apex at perpendicular height $h_T/4$ above its medial base, and four rigidity layers force all 16 apices to coincide at the centroid $G_T = (0, 0, 0)$:*

- (L_1) *Volume forces apex height $= h_T/4$.*
- (L_2) *Same-face medial-edge matching forces the apex of the corner- v_j piece on face F_i to equal the apex of the central piece on F_i .*
- (L_3) *T -edge matching propagates the apex across faces, forcing all four central apices to coincide at a common point Q .*
- (L_4) *Q is at perpendicular distance $h_T/4$ from every face; the 3-of-4 face-normal linear system has unique solution $Q = G_T$.*

Lemma 12.11 (Edge-multiset incongruence at G_T , C24v). *At $Q = G_T$, the corner piece has squared-edge multiset $\{1, 1, 2, 2, 2, 3\}$ and the central piece has squared-edge multiset $\{1, 1, 1, 2, 2, 2\}$. These multisets differ; hence the corner and central pieces are not congruent.*

Proof of conditional theorem 12.8. Combine [lemma 12.9](#), [lemma 12.10](#), and [lemma 12.11](#) with the medial-base reduction encoded in $(H_{\text{roles}}) + (H_{\text{slot}})$: the first forces tetrahedrality; the second collapses all apices to G_T ; the third produces a direct violation of piece congruence. □

Remark 12.12 (Scope of [theorem 12.7](#) and [conditional theorem 12.8](#)). In the Coxeter-pure instance $(H_{\text{roles}}) + (H_{\text{slot}})$ are automatic, so [theorem 12.7](#) is the L3 (unconditional) specialisation of the L2 conditional statement [conditional theorem 12.8](#). In the general convex instance $(H_{\text{roles}}) + (H_{\text{slot}})$ are residual; their theorem-level discharge is the remaining isolated bridge lemma (the paired Star^+ transport lemma of the supplementary programme), so [conditional theorem 12.8](#) is at evidence level L2.

Remark 12.13 (Convexity hypothesis). The convexity of the pieces is essential to [lemma 12.9](#): without it, multi-apex non-convex pieces are not reached by the perpendicular-bisector argument (ARG-2). Discharging convexity is natural future work; the present theorem is, as stated, in the convex category.

Remark 12.14 (Why $n = 16$?). $n = 16$ divides neither 24 nor 48. Note that closing the full $n = 16$ case *would* advance [conjecture 1.1](#), since $16 \nmid 24$ means that the conjecture predicts no dissection should exist. However, [lemma 12.9](#) closes only the *all-one-face residual* configuration at $n = 16$, not the entire case. This particular configuration was historically the most resistant residual of the role-distribution table in the programme described in [Section 12.2](#). The same C24 machinery is structurally portable to the other non-divisors $n \in \{20, 28, \dots\}$ and gives a template for the configurational attack on GAP B.

12.5 Certificate reduction at $n = 5$: the first non-divisor reduced to one (H-orb) residual

The C24-chain of [Section 12.4](#) reduced the largest role-distribution residual at $n = 16$ to a conditional statement. The current $n = 5$ public stack is sharper than the earlier four-hypothesis formulation. The active ledger is:

- (H-trip) is retired: it is no longer a formal input to the public theorem.
- (H-coc) is discharged directly by the orientation-bit certificate `hcoc_orientation_certificate.py`.
- (H-Qb) is discharged by the Qb certificate stack `hqb*_certificate.py`.
- (H-orb) is not discharged in blanket form. Its singleton and odd-multiplicity branches are killed by parity, and its only remaining branch is the accidental even metric residual described in [Conditional Theorem 12.16](#).

Thus the correct theorem-level claim is not an unconditional impossibility of $n = 5$, but a reduction of every putative dissection to one explicitly scoped H-orb residual. Throughout this subsection the standing embedding is (7).

Volume, face area, height of T and per-piece totals in the $n = 5$ case:

$$V_T = \frac{8}{3}, \quad A_T = 2\sqrt{3}, \quad h_T = \frac{4}{\sqrt{3}}, \quad V_{\text{piece}} = \frac{8}{15}, \quad E = \frac{8\sqrt{3}}{5},$$

where E is the per-piece total exposed (boundary) surface area, determined by the constraint $5E = 4A_T = \text{total boundary area of } T$.

Definition 12.15 (Touch count and incidence vector). For each piece P let $I(P) \subseteq \{0, 1, 2, 3\}$ be the set of T -faces on which P has positive 2-area, $t := |I(P)|$ (constant by congruence), and $k = (k_0, k_1, k_2, k_3)$ with $k_j := \#\{P : j \in I(P)\}$, so $\sum_j k_j = 5t$.

Conditional Theorem 12.16 ($n = 5$ convex reduction to the accidental-even (H-orb) residual). *Let $\mathcal{D}$ be a putative face-to-face congruent dissection of the regular tetrahedron T into 5 convex pieces. Then the public certificate stack eliminates every branch of the $n = 5$ role-distribution census except possibly the following residual:*

$$f(P) = 7, \quad t = 3, \quad \text{Sym}(P) = 1,$$

and the four interior facets of the reference piece P are accidentally congruent in an even metric pattern (4) or (2, 2), without being promoted to genuine $\text{Sym}(P)$ -orbits by orbit transport.

Equivalently, $n = 5$ is reduced to the single H-orb accidental-even residual. No unconditional impossibility of a 5-piece convex dissection is claimed at this stage. The certificate ledger is:

- (H-trip) is retired and is not used as a formal hypothesis;
- (H-coc) is closed by `hcoc_orientation_certificate.py`, which fixes the orientation bit in the K_5 triangle system;
- (H-Qb) is closed by `hqb_reduction_certificate.py`, `hqb_perface_tiling_certificate.py`, `hqb_asymmetric_caseI_certificate.py`, `hqb_asymmetric_caseII_certificate.py`, and `hqb_opp_certificate.py`;
- the singleton and odd-multiplicity (H-orb) branches are closed by `horb_parity_certificate.py`;
- `horb_residual_scope.py` and `horb_residual_combinatorial.py` identify the remaining branch as genuinely metric rather than a bookkeeping artefact.

The synthesis certificate is `phase26e_synthesis_theorem_26_1.py`, whose active outcome is “ $n = 5$ reduced to a single H-orb accidental even residual.”

The proof of the reduction occupies the remainder of [Section 12.5](#). It proceeds by a role-distribution census (Step 1 below), exact eliminations for $t = 1, 2, 4$ and the canonical $t = 3$ residual, and then the $f = 5$ and $f = 7$ ledgers that isolate the single accidental-even H-orb branch.

Step 1: role-distribution census

Since each T -face must be covered by the boundary of the dissection, $k_j \geq 1$ for every j . Each piece contributes area at most E to any single T -face, so $A_T \leq k_j \cdot E$, i.e.

$$k_j \geq \left\lceil \frac{A_T}{E} \right\rceil = \left\lceil \frac{5}{4} \right\rceil = 2. \quad (8)$$

Combined with $\sum_j k_j = 5t$ the lower bound (8) forces $5t \geq 8$, hence $t \geq 2$ (and $t \leq 4$ trivially). In particular, the $n = 5$ analogue of the $n = 16$ all-one-face residual ($t = 1$, $k = (5, 0, 0, 0)$) is *combinatorially* impossible for two independent reasons: three T -faces would be uncovered (contradicting ∂T -coverage), and $5E = 8\sqrt{3} = 4A_T$, four times the single-face capacity.

After quotienting by S_4 acting on T -faces exactly *seven* residuals survive the filter (8):

t	k -orbit representative (sorted desc)	$\sum k_j$	Per- T -face uniform-share a^2
2	(4, 2, 2, 2)	10	3/4, 3, 3, 3
2	(3, 3, 2, 2)	10	4/3, 4/3, 3, 3
3	(5, 5, 3, 2)	15	12/25, 12/25, 4/3, 3
3	(5, 4, 4, 2)	15	12/25, 3/4, 3/4, 3
3	(5, 4, 3, 3)	15	12/25, 3/4, 4/3, 4/3
3	(4, 4, 4, 3)	15	3/4, 3/4, 3/4, 4/3
4	(5, 5, 5, 5)	20	12/25, 12/25, 12/25, 12/25

Step 2: the $t = 4$ residual (5, 5, 5, 5)

In this residual every piece touches all four T -faces ($t = 4$). Write (m_1, m_2, m_3, m_4) for the canonical boundary-area multiset of the common piece shape; by congruence $M := \sum_j m_j$ is the same for every piece and $5M = 4A_T$, so $M = 4A_T/5$. We make *no* assumption that the m_j are equal.

Lemma 12.17 (Regular-tetrahedron barycentric distance sum). *For any point $p \in T$,*

$$\sum_{j=0}^3 d(p, \text{aff}(F_j)) = h_T,$$

where $F_0, \dots, F_3$ are the four faces of T .

Proof. Standard application of $V_T = (1/3) \sum_j \text{area}(F_j) \cdot d(p, \text{aff}(F_j))$ with all $\text{area}(F_j) = A_T$ and $V_T = (1/3)A_T h_T$. $\square$

Lemma 12.18 (Centroid pyramid decomposition on the closure). *For any convex polytope $P \subset \mathbb{R}^3$ and any point $q \in P$ (closure, not just interior),*

$$\text{Vol}(P) = \frac{1}{3} \sum_{F \in \mathcal{F}(P)} \text{area}(F) d(q, \text{aff}(F)).$$

Every summand is nonnegative; it vanishes if and only if $q \in \text{aff}(F)$.

Proof. Apply the divergence theorem to $v(x) = (x - q)/3$ on P . Since $\text{div } v = 1$,

$$\text{Vol}(P) = \int_P 1 \, dx = \int_{\partial P} v \cdot \hat{n} \, dA = \frac{1}{3} \sum_{F \in \mathcal{F}(P)} \text{area}(F) d_{\text{signed}}(q, \text{aff}(F)),$$

where d_{signed} is the signed distance (positive on the side of the outward normal). For $q \in P$ (convex), every supporting hyperplane has q on its inner side, so all signed distances are nonnegative and equal to the ordinary distance. $\square$

Let q_T be the incenter of T , the unique interior point with $d(q_T, \text{aff}(F_j)) = h_T/4$ for $j = 0, \dots, 3$. Since the five closed pieces cover T and $q_T \in \text{int}(T)$, at least one piece P^* satisfies $q_T \in P^*$. Apply lemma 12.18 at $q = q_T$:

$$V_{\text{piece}} = \underbrace{\frac{1}{3} \sum_{j=0}^3 m_{\sigma^*(j)} d(q_T, F_j)}_B + \underbrace{\frac{1}{3} \sum_{G \in \mathcal{F}_{\text{int}}(P^*)} \text{area}(G) d(q_T, G)}_{I \geq 0}. \quad (9)$$

The boundary term depends only on the total M :

$$B = \frac{h_T}{12} \sum_j m_{\sigma^*(j)} = \frac{h_T M}{12} = \frac{h_T}{12} \cdot \frac{4A_T}{5} = \frac{A_T h_T}{15} = \frac{V_T}{5} = V_{\text{piece}}.$$

Hence $I = 0$: every interior facet G of P^* satisfies $d(q_T, \text{aff}(G)) = 0$, i.e. $q_T \in \text{aff}(G)$.

Non-degenerate case. If $q_T \in \text{int}(P^*)$ then $d(q_T, \text{aff}(G)) > 0$ for every facet G , so P^* has no interior facets: $f(P^*) = 4$. By congruence every piece has $f = 4$, contradicting face-to-face (no shared interior facet).

It remains to exclude the degenerate case $q_T \in \partial P^*$. Every point of a convex polytope belongs to the relative interior of a unique face σ ; write $q_T \in \text{relint}(\sigma)$. The four boundary facets of P^* lie on T -face planes at distance $h_T/4 > 0$ from q_T , so σ is not a boundary facet and lies in the interior skeleton. We distinguish three cases by $\dim \sigma$.

Lemma 12.19 (Degenerate face case). *σ cannot be an interior facet of P^* .*

Proof. Suppose $q_T \in \text{relint}(G)$ for an interior facet G . Then $\text{aff}(G)$ is the supporting hyperplane of P^* at q_T . For any other facet $G' \neq G$, $q_T \in \text{aff}(G')$ forces $q_T \in G'$ (since $\text{aff}(G')$ supports P^* and $q_T \in P^*$), placing $q_T \in G \cap G'$ —a proper face of G , contradicting $q_T \in \text{relint}(G)$. Thus P^* has exactly one interior facet: $f - 4 = 1$. Double-counting interior facet-instances gives $5(f - 4)$ even, so $f - 4$ is even. Contradiction. $\square$

Lemma 12.20 (Degenerate edge case). *σ cannot be an interior edge of P^* .*

Proof. Suppose $q_T \in \text{relint}(e)$ for an interior edge e of P^* . From $I = 0$, every interior facet G satisfies $q_T \in \text{aff}(G)$; since $\text{aff}(G)$ is a supporting hyperplane of P^* and $q_T \in P^*$, we obtain $q_T \in G$. In the face lattice of a convex polytope, $\text{relint}(e) \cap G \neq \emptyset$ forces $e \subseteq G$, so every interior facet contains e . An edge of a 3-polytope is contained in exactly two facets, giving exactly two interior facets; the four boundary facet-planes, at distance $h_T/4 > 0$ from q_T , confirm that neither facet containing e is boundary. Hence $f = 6$; by congruence every piece has $f = 6$.

Each interior facet is shared with a unique neighbour, so the adjacency graph is 2-regular on 5 vertices, hence the 5-cycle C_5 . The facet G shared between P^* and a neighbour P' satisfies $q_T \in e \subset G$, so $q_T \in P'$. Iterating around C_5 shows $q_T \in P_k$ for all k .

Applying the $I = 0$ argument to each piece, all $10 = 5 \times 2$ interior facets lie in planes through q_T . The spherical link at q_T is a tiling of S^2 by five congruent digons (each piece contributes two interior-facet arcs). A digon is bounded by two great-circle arcs sharing the same pair of antipodal vertices; each spherical digon is therefore determined by a unique such pair. Adjacent digons sharing an edge share that pair, and since the edge-adjacency graph of the tiling is connected, all five digons share the same antipodal pair; hence the tiling is a hosohedron. Equivalently, all interior-facet planes contain a single line L through q_T .

Let $E = L \cap T$ and let H be the plane through the midpoint m of E perpendicular to L . Set $K = T \cap H$ (a convex polygon with at most four sides). Since L is contained in every interior-facet plane, the segment E lies in every piece and is the intersection of the two interior facets of each P_k . We claim that any isometry $\phi_k: P^* \rightarrow P_k$ maps boundary facets to boundary facets (and hence interior facets to interior facets). The four boundary facets of P^* lie on the four face-planes of T , whose six pairwise dihedral angles all equal $\arccos(1/3)$. An isometry preserves dihedral angles, so the images form four facets of P_k whose supporting planes also have all pairwise angles $\arccos(1/3)$. Among the six facet-planes of P_k , the two interior planes meet at $360^\circ/5 = 72^\circ \neq \arccos(1/3)$, so both interior planes cannot appear in such a quadruple. If exactly one interior plane were included, it would make angle $\arccos(1/3)$ with the remaining three face-planes of T ; but the four face-normals of a regular tetrahedron satisfy $n_i \cdot n_j = -\frac{1}{3}$ for $i \neq j$, so a unit vector u with $|u \cdot n_i| = \frac{1}{3}$ for $i = 1, 2, 3$ satisfies $u = \pm n_4$, forcing the interior plane to coincide with a face-plane of T —impossible since interior-facet planes pass through $q_T \in \text{int}(T)$. Thus ϕ_k maps boundary facets to boundary facets.

Consequently ϕ_k sends the two interior facets of P^* to those of P_k and their common edge E to the corresponding edge of P_k , which is again E . In particular ϕ_k fixes the midpoint m and preserves H , so the five cross-sections $S_k = P_k \cap H$ are congruent convex regions, each a sector with apex m and angular width 72° .

By congruence the five radial edges from m to the division points $d_1, \dots, d_5 \in \partial K$ all have a common length r , so the d_k lie on a circle C of radius r centred at m . The points $d_1, \dots, d_5$ split ∂K into five arcs with pairwise disjoint interiors. Each vertex of K lies in the interior of at most one arc, and K has at most four vertices, so at least one arc contains no vertex of K ; that arc is a single straight segment. Since all five outer arcs are congruent (as images of one another under the ϕ_k), every arc is a straight segment lying on one edge of K . With five straight arcs and at most four edges, two consecutive arcs share the same edge, placing three of the d_k on one line. But a line meets a circle in at most two points. Contradiction. $\square$

Lemma 12.21 (Degenerate vertex case). *σ cannot be a vertex of P^* .*

Proof. If q_T is a vertex of P^* , at least three facets meet there. Since the four boundary facet-planes are at distance $h_T/4 > 0$, all facets at q_T are interior. Each interior facet is shared with exactly one of the other four pieces, and two convex pieces in a face-to-face dissection share at most one facet (their intersection is convex, hence a single face), so P^* has at most four interior facets. Combined with at least three and the parity constraint $f - 4$ even, exactly four interior facets remain, giving $f = 8$. By congruence every piece has $f = 8$.

Each piece has four interior facets, giving a 4-regular adjacency graph on five vertices, i.e. K_5 . Every pair shares an interior facet through q_T , so all five pieces meet at q_T . Reapplying the $I = 0$ argument to each piece, every interior facet of every piece passes through q_T .

Intersect with a small sphere S_ε centred at q_T . The five links $\Lambda_k = P_k \cap S_\varepsilon$ are congruent spherical quadrilaterals (4 edges each) tiling S^2 edge-to-edge. By Euler's formula $F_s = 5$, $E_s = \frac{1}{2} \cdot 4 \cdot 5 = 10$, $V_s = 2 + E_s - F_s = 7$. In a face-to-face dissection every interior edge of the cell complex is shared by at least three cells (distinct neighbours in K_5), so every vertex of this spherical tiling has degree ≥ 3 , giving $2E_s \geq 3V_s$. But $20 < 21$. Contradiction. $\square$

[Lemmas 12.19 to 12.21](#) exhaust every degenerate position of q_T on ∂P^* . Together with the non-degenerate case, the $(5, 5, 5, 5)$ residual is UNSAT.

Step 3: the $t \in \{2, 3\}$ residuals

Rescale areas to integer units, $A_T = 60$, $E = 48$ (natural units of $\sqrt{3}/30$). By piece congruence every piece carries the *same* canonical footprint multiset $(m_1, \dots, m_t)$ with $\sum m_i = E = 48$; for each T -face j with $k_j > 0$ there must exist a k_j -multisubset of the multiset summing to $A_T = 60$.

$t = 2$ residuals. Both $(4, 2, 2, 2)$ and $(3, 3, 2, 2)$ contain a face with $k_j = 2$; the pair-sum requirement $m_a + m_b = 60$ combined with $m_1 + m_2 = 48$ forces $(m_1, m_2) = (18, 30)$ uniquely (taking $2 \cdot 30 = 60$). Then:

- for $(4, 2, 2, 2)$ the $k_0 = 4$ face needs a 4-multisubset of $\{18, 30\}$ summing to 60, but the achievable sums are $\{72, 84, 96, 108, 120\}$, missing 60;
- for $(3, 3, 2, 2)$ the $k_0 = 3$ face needs a 3-multisubset summing to 60, but the achievable sums are $\{54, 66, 78, 90\}$, missing 60.

Both residuals are UNSAT at the canonical-multiset level.

$t = 3$ residuals. Exhaustive enumeration over ordered triples (m_1, m_2, m_3) with $\sum m_i = 48$ and $m_i \geq 1$, combined with the per- T -face contribution-sum test and a brute-force search for per-piece assignments, gives:

Residual	Multisets passing contribution test	Consistent assignments	Status
$(5, 5, 3, 2)$	0	–	UNSAT at canonical-multiset level
$(5, 4, 4, 2)$	1 (namely $(6, 12, 30)$)	24	combinatorially SAT
$(5, 4, 3, 3)$	5	20	combinatorially SAT
$(4, 4, 4, 3)$	8	54	combinatorially SAT

Only $(5, 5, 3, 2)$ closes at the canonical-multiset level. Pooling the contribution-sum-feasible area multisets of the other three residuals yields the finite set

$$\mathcal{M} = \{ (6, 12, 30), (6, 15, 27), (6, 18, 24), (12, 12, 24), (12, 16, 20) \} \quad (10)$$

of five integer triples of distinct T -face footprints (units of $\sqrt{3}/30$). We close each of them by a topological reduction.

Lemma 12.22 (Topological parity for $t = 3$ pieces). *Let $P_1, \dots, P_5$ be a face-to-face congruent dissection of T with $t(P) = 3$ for every piece. Then each piece has $f(P) \in \{5, 7\}$ facets.*

Proof. Let $f = f(P)$. We first count boundary facets. A convex polytope has *at most* one facet per supporting hyperplane (two coplanar facets would either coincide, contradicting the definition of a facet, or leave a lower-dimensional gap between them, contradicting convexity).

Hence P has at most one facet on each of the $t = 3$ touched T -faces. On the other hand, $t(P) = 3$ means P has positive 2-area on three T -faces, so it contributes at least one facet to each of them. Thus P has exactly 3 boundary facets and $f - 3$ interior facets.

Next we count interior incidences. Two distinct convex pieces P_i, P_j can share at most one facet: in a face-to-face dissection any two pieces have convex intersection equal to a (possibly empty) face of each, and a single convex face cannot equal two different facets of either piece. Consequently each interior facet of P_i is shared with a *unique* other piece, and is counted twice in the total piece–facet incidence count on interior facets. The total is $5(f - 3)$, hence even. Since 5 is odd, $f - 3$ is even, i.e. f is odd.

Finally, convexity forces $f \geq 4$, and the one-facet-per-neighbour bound from the previous paragraph gives $f - 3 \leq n - 1 = 4$, so $f \in \{5, 7\}$. $\square$

Remark 12.23 (Tetrahedral sanity check). The excluded $f = 4$ case can be ruled out a second time by elementary arithmetic, which we record as an independent consistency check. A tetrahedral piece P with $t = 3$ must have all three boundary facets meeting at a single vertex v_P of T (the three T -face planes meet only at the four vertices of T); the three edges of P issuing from v_P therefore lie along the three edges of T at v_P , with pairwise angles 60° . Writing a, b, c for the three edge lengths, the boundary triangles at v_P have areas

$$m_1 = \frac{bc\sqrt{3}}{4}, \quad m_2 = \frac{ac\sqrt{3}}{4}, \quad m_3 = \frac{ab\sqrt{3}}{4},$$

and the volume of the resulting tetrahedron evaluates (Gram determinant) to $V(P) = abc/(6\sqrt{2})$. Eliminating a, b, c yields

$$m_1 m_2 m_3 = \frac{3\sqrt{3}}{64} (abc)^2 = \frac{27\sqrt{3}}{8} V(P)^2 = \frac{27\sqrt{3}}{8} \cdot \frac{V_T^2}{25} = \frac{A_T^3}{25} = 8640 \quad (11)$$

in the integer units of (10) (with $A_T = 60$, using $V_T = (1/3)A_T h_T$ and $h_T^2 = 2A_T/3$). On the other hand, by AM–GM any triple (m_1, m_2, m_3) with $m_1 + m_2 + m_3 = 48$ satisfies $m_1 m_2 m_3 \leq (48/3)^3 = 4096 < 8640$. The tetrahedral case is therefore arithmetically infeasible, confirming the topological exclusion of Lemma 12.22.

Lemma 12.24 (Weak Congruence Transport). *Let $P_1 = \gamma_1(P), \dots, P_5 = \gamma_5(P)$ be congruent copies of a reference polytope P under rigid motions $\gamma_i \in \text{Isom}(\mathbb{R}^3)$, and suppose pieces P_i and P_j share a facet S_{ij} which is the image $\gamma_i(F_a)$ of abstract facet F_a of P and also the image $\gamma_j(F_b)$ of abstract facet F_b of P . Then the composition $\tau_{ij} := \gamma_j^{-1}\gamma_i \in \text{Isom}(\mathbb{R}^3)$ restricts to an isometry $\tau_{ij}|_{F_a} : F_a \rightarrow F_b$ of polygons. In particular F_a and F_b share*

- (i) *combinatorial type: same vertex count, same edge count;*
- (ii) *metric invariants: same multiset of edge lengths, same multiset of interior angles, and the same 2-area.*

Proof. $\gamma_i(F_a) = S_{ij} = \gamma_j(F_b)$ yields $\tau_{ij}(F_a) = \gamma_j^{-1}(S_{ij}) = F_b$. Restricted to F_a , the rigid motion τ_{ij} of $\mathbb{R}^3$ is an isometry of polygons $F_a \rightarrow F_b$; combinatorial data and metric invariants of F_a are thereby transferred to F_b . $\square$

Remark 12.25 (The strong claim $\tau_{ij} \in \text{Sym}(P)$ does not hold in general). It is *incorrect* to conclude that τ_{ij} belongs to $\text{Sym}(P)$: while $\tau_{ij}(F_a) = F_b$ as polygons, one has $\tau_{ij}(P) = \gamma_j^{-1}(P_i)$, i.e. the image of P lies on the side of F_b opposite P , so $\tau_{ij}(P) = P$ would force P to be reflection-symmetric across the plane of F_b , a rare accident. (Equivalently, $\tau_{ij} \in \text{Sym}(P)$ would require $P_i = P_j$ as subsets of $\mathbb{R}^3$, impossible for distinct pieces.) The correct statement is Lemma 12.24 above, which we refer to as the *weak* Congruence Transport Lemma; it already suffices to close the $f = 5$ case. The $f = 7$ argument below uses the following orbit-level corollary.

Lemma 12.26 (Orbit transport). *Keep the hypotheses of Lemma 12.24 and assume moreover the following orbit-detection property of the reference tile P :*

(H-orb) *Any two facets of P with equal edge-length and interior-angle multisets lie in a common $\text{Sym}(P)$ -orbit.*

Then F_a and F_b lie in a common $\text{Sym}(P)$ -orbit on the facet set of P , and there exists $g_{ij} \in \text{Sym}(P)$ with $g_{ij}(F_a) = F_b$.

Proof. By Lemma 12.24, F_a and F_b have equal edge-length and interior-angle multisets; (H-orb) gives the orbit claim, and the transitive action on the orbit supplies g_{ij} . $\square$

Remark 12.27 (Current status of (H-orb)). The blanket orbit-detection statement (H-orb) is no longer used as an undischarged standing hypothesis. The weak Congruence Transport Lemma gives metric equality of paired facets. The certificate `horb_parity_certificate.py` then kills every odd-multiplicity interior-facet pattern, including the singleton patterns that earlier drafts handled through (H-orb). The only surviving patterns are the even ones, (4) and (2, 2). When these are genuine $\text{Sym}(P)$ -orbits, the orientation certificate and Lemma 12.28 close the branch. What remains is precisely the accidental case with $\text{Sym}(P) = 1$, where equal facet metrics do not come from a symmetry of P ; this is the open H-orb residual recorded in `horb_residual_scope.py`.

Lemma 12.28 (K_5 triangle cycle-consistency). *Let $P_1 = \gamma_1(P), \dots, P_5 = \gamma_5(P)$ be a face-to-face congruent dissection with $f(P) = 7$, and assume the reference tile P has $\text{Sym}(P) = \langle \sigma \rangle \cong C_2$ whose induced action on the four interior facets of P has orbit structure (2, 2). Assume also hypotheses (H-orb) of Lemma 12.26 and the following triangle cocycle:*

(H-coc) *For every triple $\{i, j, k\} \subset \{1, \dots, 5\}$, the chosen orbit-transport elements $g_{ij}, g_{jk}, g_{ki} \in \text{Sym}(P)$ from Lemma 12.26 satisfy $g_{ij} \cdot g_{jk} \cdot g_{ki} = \text{id}$ in $\text{Sym}(P)$.*

Then no such dissection exists.

Remark 12.29 (Historical (H-trip) route to (H-coc)). Earlier drafts recorded a developing-map route intended to derive (H-coc) from the topological statement (H-trip). That route is no longer an active dependency of Conditional Theorem 12.16: the orientation-bit certificate `hcoc_orientation_certificate.py` discharges (H-coc) directly. We retain the label (H-trip) only as historical motivation for why the cocycle condition was isolated; it is retired from the formal public ledger.

Remark 12.30 (Current logical chain). The active $f = 7$ chain no longer depends on four standing hypotheses. The orientation-bit certificate fixes the H-coc sign obstruction directly; H-trip is retired. Parity closes the singleton and odd H-orb branches. The genuine (2, 2) orbit branch is closed by the H-coc orientation obstruction, while `horb_residual_scope.py` records the one branch not closed by these soft tools: accidental even metric congruence with $\text{Sym}(P) = 1$. The synthesis result file `phase26e_synthesis_theorem_26_1_results.json` records this split via `closed_modules`, `retired_modules`, and `remaining_modules`.

Proof of Lemma 12.28. By Lemma 12.26 applied edgewise (using (H-orb)), each shared interior facet S_{ij} determines an element $g_{ij} \in \text{Sym}(P)$; by the C_2 assumption we have $g_{ij} \in \{\text{id}, \sigma\}$, encoded by a bit $b_{ij} := \text{bit}(g_{ij}) \in \{0, 1\}$.

Step 1: K_5 decomposes into two edge-disjoint 5-cycles. Let A be the set of K_5 -edges whose shared facet lies in the first $\text{Sym}(P)$ -orbit (size 2) of interior facets of P , and B the set of edges in the second. Each piece P_i has 4 interior facets, two per orbit; therefore i has A -degree 2 and B -degree 2. Hence A and B are two 2-regular graphs on 5 vertices, edge-disjoint, with $|A| = |B| = 5$. The unique 2-regular simple graph on 5 vertices is the 5-cycle C_5 (any 2-regular

graph is a disjoint union of cycles, and 5 is prime so it must be a single cycle), so $A = B = C_5$ as graphs and $A \sqcup B = K_5$.

Step 2: bit parametrisation. At each piece P_i the two A -slots carry the two A -orbit elements $\{F_A, \sigma F_A\}$ of P *exactly once each* (distinct facets carry distinct abstract labels). Orient the A -cycle by a fixed cyclic order on $\{1, \dots, 5\}$ and, at each i , call “forward A -slot” the slot pointing along this orientation and “backward A -slot” the remaining A -slot; let $x_i \in \{0, 1\}$ record whether the forward slot at i carries F_A ($x_i = 0$) or σF_A ($x_i = 1$). Since each P_i uses each A -orbit element exactly once, the backward A -slot at i then necessarily carries the *complementary* element $\sigma^{1+x_i} F_A$. On the directed A -edge $i \rightarrow j$ the shared interior facet S_{ij} is the forward A -slot of i and the backward A -slot of j , yielding

- forward label at i : $\sigma^{x_i} F_A$;
- backward label at j (same physical facet S_{ij} , viewed from j): $\sigma^{1+x_j} F_A$, by the complementarity observation above applied at j .

The orbit-transport equation $g_{ij}(\sigma^{x_i} F_A) = \sigma^{1+x_j} F_A$ in $\text{Sym}(P) = C_2$ gives

$$g_{ij} = \sigma^{1+x_i+x_j}, \quad \text{i.e.} \quad b_{ij} \equiv 1 + x_i + x_j \pmod{2} \quad \text{on } A\text{-edges,}$$

and analogously $b_{ij} \equiv 1 + y_i + y_j \pmod{2}$ on B -edges using y_i for the forward B -slot. (The “+1” records that the forward slot at i and the backward slot at j necessarily use *different* orbit elements, since each piece uses each orbit element exactly once per orbit.)

Step 3: triangle reduction. Recall from Step 1 that A and B are each isomorphic to C_5 (the 5-cycle on 5 vertices). A 5-cycle is *triangle-free* (its girth equals 5): no three vertices of C_5 form a triangle in A alone, and likewise in B alone. Hence every triangle of $K_5 = A \sqcup B$ has at least one A -edge and at least one B -edge; the two admissible A/B -splits are therefore $(|A|, |B|) = (1, 2)$ or $(2, 1)$. In either case exactly two of its three vertices are incident to *both* an A -edge and a B -edge within the triangle – call them the *mixed* vertices $\{i', j'\}$. Summing $b_{ij} + b_{jk} + b_{ki} \pmod{2}$ collapses the three “+1” terms to 1 and the six “ x_r and y_r ” terms telescope, each vertex $r \in \{i, j, k\}$ contributing $x_r + y_r = z_r$ iff r is mixed and 0 otherwise. Thus

$$b_{ij} + b_{jk} + b_{ki} \equiv 1 + z_{i'} + z_{j'} \pmod{2}, \quad z_r := x_r + y_r \pmod{2}.$$

Step 4: cocycle $\Rightarrow$ pigeon-hole. (H-coc) forces the left-hand side to vanish for every triangle, i.e. $z_{i'} + z_{j'} \equiv 1 \pmod{2}$ for every mixed pair. An elementary check (carried out in `phase26c_t3_f7_exclusion.py`) shows that as $\{i, j, k\}$ ranges over the ten triangles of K_5 , the mixed-pair $\{i', j'\}$ ranges over every unordered pair of $\{1, \dots, 5\}$; consequently $z_r \neq z_s$ for every $r \neq s$, forcing the five bits z_r to be pairwise distinct – impossible in the two-element set $\{0, 1\}$. This contradicts the existence of the dissection.

Finally, a brute-force enumeration over all $2^{10} = 1024$ parametric assignments $((x_i)_{i=1}^5, (y_i)_{i=1}^5)$ (see `phase26c_t3_f7_exclusion.py`) confirms that no (x, y) produces a bit vector b satisfying all ten triangle cocycle equations. $\square$

Closing the $f = 5$ sub-case. A convex 3-polytope with $f = 5$ is either a quadrilateral pyramid (four triangles + one quadrilateral) or a triangular prism (two triangles + three quadrilaterals). A convex polygonal facet of a convex 3-polytope contained in T that lies on ∂T necessarily lies on a *single* T -face plane (by convex support of T). Each $t = 3$ piece must place *three* of its five facets on ∂T distributed over three distinct T -face planes (three by the hypothesis $t = 3$; distinct because each of the three touched T -faces receives positive 2-area from the piece), giving the five combinatorial sub-cases enumerated below. Three of the five are excluded unconditionally by elementary metric arguments. Earlier versions isolated the remaining two sub-cases – **Qb-adj** and **Qb-opp** – as the explicit geometric bridge (H-Qb). The current public certificate stack discharges that bridge, as recorded below and in `hqb_certificate.py`.

- *Pyramid with three lateral triangles on ∂T ; prism with one triangle and two quadrilaterals on ∂T* (unconditional closure). In both configurations the two interior facets have *mis-matched* edge-counts: one of them is a triangle and the other is a quadrilateral. By the weak [Lemma 12.24](#), at each shared interior facet the two incident pieces carry congruent abstract polygons, hence in particular equal edge-count. Thus the five triangle-slot interior facets (one per piece) pair among the five pieces into a perfect matching on the 5 vertex-labels $\{1, \dots, 5\}$, as do the five quadrilateral-slot interior facets; a perfect matching on an odd number of vertices is impossible, so these two sub-cases are excluded.
- *Pyramid with its quadrilateral base on ∂T* (case **Qb**; closed by certificates). Label the pyramid as $(b_1, b_2, b_3, b_4; a)$ with $b_1, \dots, b_4$ the four base vertices cyclically around the quadrilateral on a T -face plane T_α and a the apex. The two remaining lateral triangles may be adjacent (**Qb-adj**) or opposite (**Qb-opp**). Earlier versions treated their exclusion as the standing hypothesis (**H-Qb**). The present public stack discharges this branch by the exact certificates `hqb_reduction_certificate.py`, `hqb_perface_tiling_certificate.py`, `hqb_asymmetric_caseI_certificate.py`, `hqb_asymmetric_caseII_certificate.py`, and `hqb_opp_certificate.py`. Thus no $f = 5$ sub-case remains open.
- *Prism with its two triangular faces on ∂T* (unconditional closure). The two triangular faces of a prism are parallel. If both sit on T -faces, those two T -face planes must be parallel; but the four T -face planes are pairwise incident with dihedral $\arccos(1/3) \neq \pi$, so no two are parallel. Sub-case excluded.
- *Prism with its three quadrilateral faces on ∂T* (unconditional closure). The three lateral edges of a prism are pairwise parallel; if each lateral quadrilateral sits on a distinct T -face, each lateral edge lies in the intersection of two T -face planes, i.e. on a T -edge. The three T -edges containing lateral edges are pairwise incident (they meet at a common T -vertex, being the three edges issuing from the T -vertex opposite the fourth unused T -face); in particular no two of the three chosen T -edges are parallel, contradicting the prism's parallel-lateral-edges property.

Remark 12.31 (Status of (**H-Qb**)). The **Qb-adj** and **Qb-opp** families are non-empty at the single-piece level, so the old single-piece degeneracy argument remains retracted. Their exclusion is now supplied by the multi-script **H-Qb** certificate stack listed in [Conditional Theorem 12.16](#). Consequently (**H-Qb**) is not an open standing hypothesis in the current public ledger; it is a closed module whose local decomposition is still visible in `phase26c_t3_f5_exclusion.py`.

The five-sub-case closure is certified uniformly in $\mathcal{M}$ by `phase26c_t3_f5_exclusion.py` (21/21 passing tests).

Closing the $f = 7$ sub-case. If $f(P) = 7$, each piece has four interior facets and the interior adjacency graph is K_5 . The old analysis closed this branch under a blanket (**H-orb**) assumption plus (**H-coc**). The current ledger is more precise:

- odd or singleton interior-facet multiplicity patterns are impossible by the parity certificate, without invoking blanket (**H-orb**);
- genuine $(2, 2)$ orbit transport is closed by the direct **H-coc** orientation certificate together with [Lemma 12.28](#);
- the remaining case is accidental even metric congruence with $\text{Sym}(P) = 1$, in pattern (4) or $(2, 2)$.

The legacy script `phase26c_t3_f7_exclusion.py` records the older conditional closure under blanket orbit-detection assumptions. The active public synthesis is instead `phase26e_synthesis_theorem_26_1.py` (46/46 passing tests), supported by `horb_parity_certificate.py`, `horb_residual_scope.py`, and `horb_residual_combinatorial.py`.

Synthesis of the $t = 3$ step. The four $t = 3$ residuals now split as follows. The residual $(5, 5, 3, 2)$ is UNSAT at the canonical-multiset level. In the other three $t = 3$ residuals, the $f = 5$ branch is closed by the H-Qb certificates, and the $f = 7$ branch is closed except for the accidental even H-orb residual with $\text{Sym}(P) = 1$.

Conclusion of Step 3 and proof of [conditional theorem 12.16](#)

Steps 1, 2 and 3 eliminate the $t = 1$ case, both $t = 2$ residuals, the canonical $t = 3$ residual $(5, 5, 3, 2)$, the $t = 4$ residual, all $f = 5$ branches, and every non-accidental $f = 7$ branch. Therefore any putative face-to-face congruent dissection of T into 5 convex pieces must lie in the single H-orb accidental-even residual stated in [Conditional Theorem 12.16](#). This proves the reduction theorem; no unconditional $n = 5$ impossibility is asserted here. $\square$

Remark 12.32 (Convexity hypothesis for $n = 5$). Convexity is essential to the canonical-multiset argument: a non-convex piece could touch a single T -face with multiple disjoint footprints, breaking the structural claim “footprint-multiset = t nonzero values” which underlies Step 3 above.

Remark 12.33 (Comparison with $n = 16$). The $n = 5$ reduction is structurally smaller than the C24-chain at $n = 16$. At $n = 5$ the per- T -face filter kills $t = 1$ outright, the $t = 2$ and one $t = 3$ residual close by integer footprint arithmetic, and the $t = 4$ residual closes by the full-incidence volume argument. The only surviving difficulty is not combinatorial but metric: accidental even congruence among the four interior facets of an $f = 7$ piece with trivial symmetry. This identifies a much narrower target than the earlier four-hypothesis stack.

12.6 Summary of the programme

[conjecture 1.2](#) is equivalent to [conjecture 1.1](#) combined with [Corollary 7.4](#). Its current evidence level is L1+, supported by

- one L2 theorem ([lemma 12.2](#), Stabiliser Transport);
- one L1+ proposition ([proposition 12.6](#), Conditional Separation, conditional on Hypothesis H);
- one conjecture at L1 ([conjecture 12.4](#), Boundary-Normal Orbits);
- a Spherical Link Program constraining pieces-per-vertex to $k \in \{1, 3, 6, \dots\}$ within a uniform convex model under Hypothesis H;
- the dihedral rigidity programme ([Section 12.1](#)) reducing [conjecture 1.2](#) to GAPs A, B;
- the $n = 16$ all-one-face conditional reduction ([theorem 12.7](#) and [conditional theorem 12.8](#)), L3 in the Coxeter-pure case and L2 (conditional on $(H_{\text{roles}}) + (H_{\text{slot}})$) in the general convex case;
- and the $n = 5$ convex reduction to one H-orb accidental-even residual ([conditional theorem 12.16](#)), the first non-divisor of 24 reduced in the convex category to a single named open branch. The current public stack discharges (H-coc) and (H-Qb), retires (H-trip), and kills H-orb singleton/odd patterns by parity; the remaining task is the metric accidental-even branch with $f(P) = 7$, $t = 3$, and $\text{Sym}(P) = 1$.

13 Further questions

Question 13.1 (Hinged dissections). Can any of the eight known dissections be realised as a physical hinged mechanism? Edge-hinges (1-DOF) are natural candidates for S_2 and S_4 ,

vertex-hinges (3-DOF) for the dense configurations S_{24} . No results are known in this direction. Demaine’s universality theorems on scissors-congruent polyhedra [5] apply to pairs of polyhedra, not to partitions of a single polyhedron; their adaptation to the present setting is open.

Question 13.2 (Higher-dimensional extension). For the regular d -simplex with symmetry group S_{d+1} of order $(d+1)!$, does n admit a congruent dissection if and only if $n \mid (d+1)!$? The answer is *no* for $d = 2$: the equilateral triangle admits congruent dissection into any $n = k^2$ pieces via self-similar subdivision. The tetrahedron ($d = 3$) may be the critical dimension at which the Dehn invariant first constrains the admissible n ; this should be a theorem for $d = 3$, with modifications for $d \geq 4$.

Question 13.3 (Burnside ring extensions). The Burnside ring framework of Section 9 opens questions in *equivariant K-theory* of simplicial complexes and connections to the representation ring $R(T_d)$. The relationship between the mark table and the character table of S_4 may encode deeper number-theoretic structure.

A Computational reproduction sketch

All metric, Dehn, stabiliser, MILP and SAT-style certificates cited in the paper are produced by exact rational arithmetic using Python’s built-in `Fractions` module. We include below a minimal self-contained reference implementation for the Dehn-invariant verification at $n = 24$ (orthoscheme). The full test suites for [conditional theorem 12.8](#) and [conditional theorem 12.16](#) are available in the public repository and are summarised by the data and code capsule described in [Section 1](#).

Listing 1: Minimal SymPy verification of the Dehn scaling $D(S_{24}) = D(T)/24$ for the orthoscheme piece.

```

1 import sympy as sp
2 from sympy import Rational, Point3D, sqrt
3 # Regular tetrahedron T with edge length 1
4 A = Point3D(0, 0, 0)
5 B = Point3D(1, 0, 0)
6 C = Point3D(Rational(1, 2), sqrt(3)/2, 0)
7 D = Point3D(Rational(1, 2), sqrt(3)/6, sqrt(6)/3)
8 # Canonical orthoscheme piece S_24
9 M_AB = (A + B) / 2
10 G_ABC = (A + B + C) / 3
11 G_T = (A + B + C + D) / 4
12 S24 = [A, M_AB, G_ABC, G_T]
13 # Edge enumeration, dihedral-angle computation via dot products of
14 # outward face normals, and Dehn tensor product in R (x)_Z (R/pi Q)
15 # follow directly; total runtime < 1 s.
```

B Evidence-level glossary

- L0** *Heuristic*. A claim supported by plausibility or partial checks but without a formal proof attempt.
- L1** *Partial evidence*. A claim supported by substantial evidence (e.g. computational enumeration up to a fixed bound) but lacking a closed proof.
- L1+** *Strong partial evidence*. A claim supported by a partial proof together with the explicit identification of the remaining residual gap(s); typically a reduction to a named sub-lemma.
- L2** *Conditional theorem*. A rigorously proved implication whose conclusion depends on an explicit, named auxiliary hypothesis (e.g. convexity, face-to-face, or a named hypothesis H).

- L2+** *Finite-check conditional theorem.* Like L2, but each standing auxiliary hypothesis is a finite, explicitly verifiable condition on a candidate configuration (e.g. a cocycle identity, an orbit table, or a single-piece combinatorial type exclusion).
- L3** *Unconditional theorem.* A fully proved statement with no auxiliary hypothesis beyond those of the ambient setting (convex face-to-face congruent dissection).

C The 24-cell and quaternionic action

This appendix provides a *secondary* algebraic viewpoint. The core results of the paper (Section 3–Section 9, Section 10–Section 11) do not depend on it; it serves as a partial independent verification of the dissection inventory through the quaternionic / 24-cell correspondence, and may be skipped on first reading.

C.1 The binary tetrahedral group $2T$

The binary tetrahedral group $2T \subset S^3 \subset \mathbb{H}$ consists of 24 unit quaternions: the 8 *type-1* units $\{\pm 1, \pm i, \pm j, \pm k\}$ and the 16 *type-2* units $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$. These are precisely the 24 *Hurwitz units*, the vertices of the regular 24-cell inscribed in S^3 . The group has seven conjugacy classes of sizes $(1, 6, 1, 4, 4, 4, 4)$. Its centre is $\{\pm 1\}$, and

$$1 \longrightarrow \{\pm 1\} \longrightarrow 2T \xrightarrow{\pi} A_4 \longrightarrow 1 \quad (12)$$

is exact.

C.2 Triple bijection

The quaternionic conjugation action $v \mapsto qv\bar{q}$ on pure quaternions v yields a canonical triple bijection

$$\underbrace{24 \text{ Hurwitz units}}_{2T} \longleftrightarrow \underbrace{24 \text{ permutations}}_{T_d \cong S_4} \longleftrightarrow \underbrace{24 \text{ orthoschemes}}_{\text{flags}(T)}. \quad (13)$$

Proper rotations give $24 \twoheadrightarrow 12$ even permutations (A_4) with kernel $\{\pm 1\}$; composition with a fixed involution σ recovers the 12 odd permutations ($S_4 \setminus A_4$); and each of the 24 orthoschemes in the barycentric subdivision of T is hit exactly once.

C.3 Subgroup lattice and the $n = 2$ gap

The subgroup lattice of $2T$ maps under π to the subgroup lattice of $A_4 \leq S_4$. Seven of the eight dissection numbers are recovered directly from coset spaces $2T/H$: $\{1, 3, 4, 6, 8, 12, 24\}$ (Table 15). The missing case $n = 2$ arises because $2T$ has no subgroup of order 12; indeed π has image $A_4 \leq S_4$ of index 2, and all $|\pi(H)|$ divide 12. The $n = 2$ dissection is captured by the S_4 -based Burnside analysis (Section 9) via stabiliser A_4 ; see Table 15.

References

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Table 15: Subgroups of $2T$ and dissection matches.

$2T$ -subgroup	$ H $	$[2T : H]$	$\pi(H) \leq A_4$	$ \pi(H) $	Dissection
$\{e\}$	1	24	$\{e\}$	1	S_{24}
$\mathbb{Z}_2$ (centre)	2	12	$\{e\}$	1	S_{12}
$\mathbb{Z}_3$	3	8	$\mathbb{Z}_3$	3	S_8
$\mathbb{Z}_4$	4	6	$\mathbb{Z}_2$	2	S_6
$\mathbb{Z}_6$	6	4	$\mathbb{Z}_3$	3	S_4
Q_8	8	3	V_4	4	S_3
$2T$	24	1	A_4	12	S_1

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